An Archival ALMA Search for Unresolved Spectral-Carrier Technosignatures toward Stars within 40 pc at 114–873 GHz

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ABSTRACT

We searched 405 fine-channel archival ALMA windows toward 61 systems within 40 pc for narrowband, drifting radio carriers. They cover 47.7 GHz in 26 separate intervals; 1258 coarser windows constrain only unresolved emission. Drift-corrected spectra are compared with control positions in each field. Injection into 49 windows, transferred to the rest, gives a median single-epoch sensitivity of 1.8×10^{15} W equivalent isotropic power at ninety per cent recovery, for a carrier narrower than the instrumental channel; recovery is not confirmation. Of 49 peaks reaching 5 times the scatter at control positions, 23 coincide with catalogued molecular lines, 16 with further evidence of line emission. The other 26 are consistent with chance, against 23.8 expected from control positions ($P = 0.36$), which does not exclude a smaller added population. No carrier is detected or confirmed: a non-detection, not an exclusion. Confirmation needs a repeat at a covered frequency, which 21 systems lack, and tests only a carrier returning to its predicted frequency. The archive usefully constrains detectable carriers, but heterogeneous spectral and temporal coverage, more than depth, limits confirmation.

Subject headings: Technosignatures, Search for extraterrestrial intelligence, Submillimeter astronomy, Radio spectroscopy, Debris disks

1. INTRODUCTION

The public ALMA archive holds thousands of hours of millimetre and submillimetre observations taken for other reasons, mostly studies of dust discs and planet formation. This paper searches them for a narrowband transmitter at the position of a star in the field. No such signal was found.

Searches for *technosignatures* – detectable imprints of technology, intended or not (Sheikh 2020; Vidal et al. 2026) – have examined many thousands of stars since the first proposals (Cocconi & Morrison 1959; Drake 1961), from Project Phoenix (Tarter 2001; Backus & Project Phoenix Team 2002) to the targeted, commensal and machine-learning-assisted surveys of today (Zhang et al. 2020; Czech et al. 2021; Li et al. 2026; Johnson et al. 2023; Choza et al. 2024; Morrison et al. 2023; Tremblay & Tingay 2024; Manunza et al. 2025; Ma et al. 2023; Jacobson-Bell et al. 2025; Pardo et al. 2025). The last published accounting of that effort covers only 6×10^{-18} of the space of transmitter position, power, frequency, bandwidth, polarisation and repetition rate (Wright et al. 2018). Essentially all of it lies below ∼30 GHz, and above that the record is nearly empty. Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed 40 and 17 nearby targets at 203 GHz, each in no more than 35 MHz of a single line, 1363 times less than the union searched here, and with no search over drift and no test of persistence. Of ALMA's receiver bands only Band 3 has been searched (Mason et al. 2025). The lower Band 1 holds observations of only 4 stars of this sample, proprietary until 2027 and in channels too coarse for a carrier search.

At these frequencies interstellar dispersion and scintillation are negligible and an aperture delivers far more gain per watt. But system temperatures are higher, the atmosphere is opaque in places at both ends of the link, and the narrow beam – 25 arcsec for the 12 m benchmark of §5.3 – must be aimed at the Earth deliberately. Limits are therefore stated as EIRP, which needs no transmitter model. The targets were chosen by other programmes, so the sample supports no bound on the fraction of stars hosting a transmitter (Wright et al. 2018; Margot et al. 2023; Sheikh et al. 2025) (§2.1).

The signal sought is an *unresolved spectral carrier*: emission confined to a few channels at the stellar position, present at more than one epoch, and allowed to drift as it must for a transmitter on a rotating, orbiting body (§3.1). Broadband planetary-scale leakage (Garrett 2026) is outside the search by construction. Recurrence, not brightness, separates an interesting event from a detection (Sheikh et al. 2021). Mason et al. (2025) identified the obstacles in this band: drift rates that scale with frequency, and the molecular line forest. The method meets them with a drift search, a line catalogue against which every event is compared, and a recurrence test in a separated epoch. In place of interference rejection it compares the star with control positions inside its own primary beam, which a single dish cannot resolve, and every window was re-reduced from the raw archive (§3).

Experiments are compared here on the minimum detectable received flux $F_{\rm min} = S_{\rm min}\,\delta\nu$ at each one's own channel width, because $F_{\rm min}$ rises as $\delta\nu^{1/2}$. A limit holds only for a signal narrower than the channel that set it, and the channels searched here span 7 orders of magnitude, from 2.79 Hz to 31.2 MHz. Figure 1 draws each published threshold beside its value rescaled to a common 5σ, an arithmetic convention and not a matched false-alarm probability (values in `figures/figure_data.csv` of the data archive; White & Dey 2026). Only this survey's EIRP_{90} is calibrated with injected carriers, the step Price et al. (2020) name as missing from their limits and Margot et al. (2023) measure but do not fold into theirs. No plotted limit carries

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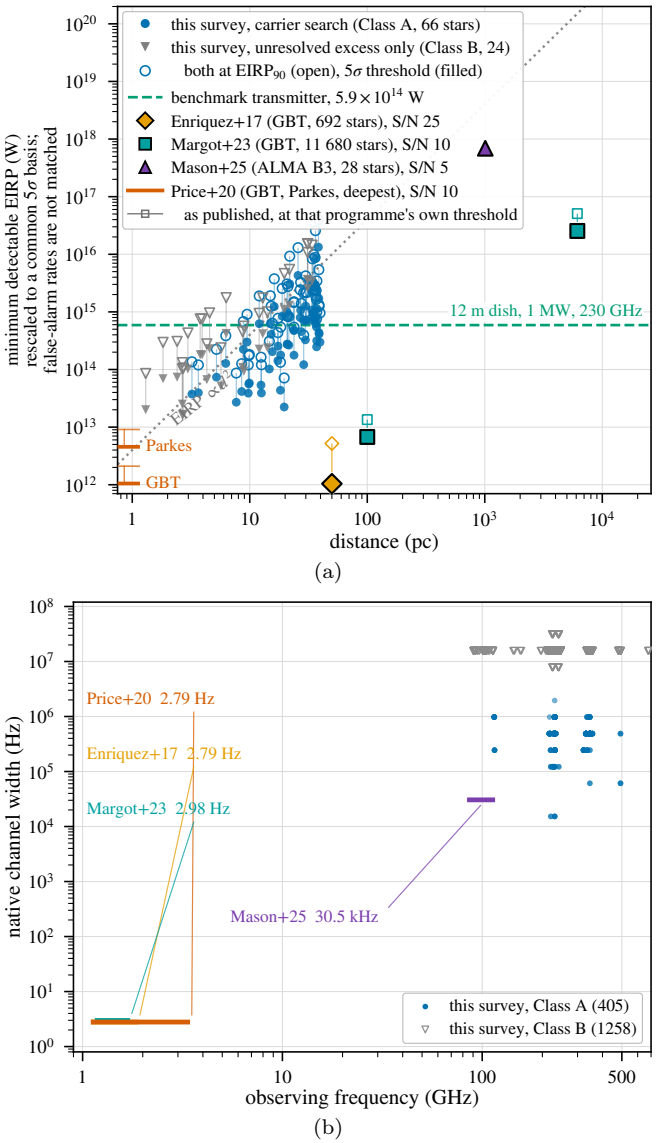


FIG. 1.— Where this survey sits among published radio technosignature searches (§1). **(a)** Minimum detectable EIRP against distance; this survey's deepest window per star – blue, the carrier search; grey, coarse channels only; filled, the bare threshold; open, EIRP_{90}. **(b)** Native channel width against sky frequency (filled, Class A; open, Class B).

a confirmation criterion. At a bare 5σ threshold Mason et al. (2025) reach a median $4.9 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ in a 30.52 kHz channel. The deepest window here outside the line mask reaches $1.2 \times 10^{-23}\,\mathrm{W\,m^{-2}}$ in 122.1 kHz, a factor 4 deeper, and the median window $3.2 \times 10^{-22}\,\mathrm{W\,m^{-2}}$ in 488 kHz, a factor 7 shallower. Per channel this survey is therefore the shallower except in its finest windows. What distinguishes it is its frequency range and its 61 systems within 40 pc, against distant stars in 4 calibrator fields.

2. THE SAMPLE

This search covers 1663 spectral windows toward 90 stars in 83 systems within 40 pc. They come from 407 public ALMA execution blocks put through one pipeline: 4.1 TB of raw data, 281 h of telescope time and 1098 h summed over windows. This *primary census* is the set over which every sensitivity, expectation and limit is computed. It holds 49 of the 51 threshold crossings this paper reports, all in quality-passing data. The remaining 2 lie in an *archival tail* of 8 blocks toward 5 census stars, searched after the census was closed, and enter no expectation. Nor do an archival extension, a reserved *hold-out* (Appendix B.1) and a control beyond 40 pc (§4.6.1). Table A1 accounts for every set, and Table A2 gives the set each count is taken over. A *system* is a bound group, counted once however many of its stars were observed.

The census is two experiments, divided on native channel width, on which both the search statistic and the completeness calibration depend. *Class A*, the primary experiment, is the 405 fine-channel windows toward 66 stars in 61 systems. Of these, 391 are channelised finely enough for the searched drift to cross a channel (§3.1), and every sensitivity and limit quoted is Class A's unless stated otherwise. *Class B*, the other 1258 windows, supports a secondary search for an unresolved spectral excess. The channelisation is bimodal, so the division is unambiguous: the widest fine channel is 1.95 MHz and the narrowest coarse one 7.81 MHz, with nothing between them.

A star enters if its Gaia DR3 parallax is at least 25 mas and measured to better than 20 per cent, a clause that never binds (the worst among the stars searched is 2.5 per cent). Its proper-motion-corrected position must fall inside the quoted field of view of a public execution block, though not necessarily at the pointing centre, and its distance is $1/\varpi$. A window is kept only where the primary-beam response at the star is at least 0.22, where the adopted Gaussian and a blocked-aperture response part by 10 per cent; no window falls below it (Appendix A.2). Of the 168 work-list entries the query returns, only 115 stars prove to lie inside a public field, and 90 yielded a searchable window. One serendipitous field star is named throughout after the ALMA source at its field centre, ALMA J153702653−33192492: it is 2MASS J15370316−3319273 (Gaia DR3 6015967739554000768), an M5 dwarf at 35.0 pc with no catalogued radial velocity.

A star with no Gaia DR3 parallax cannot enter. That removes all 22 stars in this volume brighter than second magnitude, and 257 of its 4227 qualifying Hipparcos stars in all, 198 of them fainter than Hp 4. Of these, seven lie in public fields: Sirius A, α Centauri A, Vega, Fomalhaut, α Centauri B, β Leonis and MCC 124, with 38 fine-channel and 374 coarse-channel windows (1.0 TB in 83 blocks) that a search admitting Hipparcos astrometry could add.

2.1. *The archival selection*

This sample was not designed: it is whatever ALMA programmes with other aims happened to observe among the stars Gaia DR3 catalogues, and 90 per cent of the stars searched were targeted under a debris-disc or planet-formation proposal. Table 1 measures the bias against the volume-limited Gaia DR3 census of which the searched stars are a subset ($\varpi > 0$, $\sigma_\varpi/\varpi \leq 0.1$, $1/\varpi \leq 40$ pc; 17 566 objects). Both columns are classified by BP−RP colour, with absolute G magnitude separating the degenerates.

Distance is the dominant selection effect: stars within 5 pc are over-represented by a factor 53, falling monotonically to a factor of two *under*-representation beyond

TABLE 1
Selection function of the 90 searched stars against the volume-limited Gaia DR3 census of §2.1. "Ratio": the searched over the census column fraction.

Property	Census	Searched	Ratio
Distance			
$d \leq 5$ pc	48 (0.3%)	13 (14.4%)	52.9
5–10 pc	240 (1.4%)	13 (14.4%)	10.6
10–20 pc	2069 (11.8%)	18 (20.0%)	1.7
20–30 pc	5391 (30.7%)	19 (21.1%)	0.7
30–40 pc	9818 (55.9%)	27 (30.0%)	0.5
Colour class[a]			
A	61 (0.4%)	5 (6.2%)	17.2
F	473 (2.8%)	8 (10.0%)	3.6
G	978 (5.8%)	15 (18.8%)	3.2
K	2221 (13.2%)	11 (13.8%)	1.0
M	13 082 (77.8%)	41 (51.2%)	0.7
Other axes			
Confirmed planet hosts[b]	26 (15.5%)	24 (26.7%)	1.7
White dwarfs[a]	463 (2.7%)	8 (9.1%)	3.4
Searched stars / systems	–	90 / 83	–

[a] Colour bins, not MK types: BP−RP < 0.36 for A, 0.36–0.72 for F, 0.72–0.95 for G, 0.95–1.77 for K and > 1.77 for M, where the census's own temperature–colour relation crosses 7500, 6000, 5300 and 3900 K. Degenerate: more than 4 mag below the census main sequence. The five classes are normalised over the 80 non-degenerate searched stars and the 16 815 non-degenerate census objects, the degenerate row over the 88 classified searched stars and the 17 278 classified census objects.

[b] Both columns are the same positional join, of the 168-entry ALMA-covered work list and of its searched subset, to the NASA Exoplanet Archive `pscomppars` table accessed 2026 October 4. The 24 searched hosts carry 51 confirmed planets, 22 of the 24 surviving the deuterium-burning mass limit; the Gaia census carries no planet information, so the census column in this row is the work list and not the 17 566-object Gaia census.

30 pc. Colour comes second. A stars are over-represented by about 17 and F and G stars by factors of three, because those are the stars with bright debris discs. For the A stars that factor is a floor, since the Gaia requirement itself removes stars that blue at 3.7 times its rate among the others. M dwarfs are depleted: 41 of the 90 searched stars, a factor 0.7 of their share of the census, and only 28 of them carry fine-channel coverage. Coverage is concentrated too, 285 of the 1663 windows coming from 3 systems. The selection acts hardest on the targets a designed search would pick: of the 16 sample systems within 15 pc with a confirmed planet, 6 have no Class A coverage and so no drift-resolving limit. No statement in this paper is therefore an occurrence rate for the solar neighbourhood.

2.2. *Frequency coverage and archival scope*

The Class A experiment covers $\Delta\nu_A = 47.7$ GHz of unique sky frequency, in 26 disjoint islands between 114.0 and 873.1 GHz rather than a band. With the coarse-channel windows the coverage is 118.1 GHz in 46 intervals, of which 113.1 GHz survives the molecular-line mask of §3.2. Fig. 2 shows the distribution: a transmitter between the islands was never observable here, whatever its power. Nor is the archive exhausted: 177 further blocks inside the same query were unprocessed when the census was frozen, 18 of them (237 GB) because the archive holds their raw visibilities but never pipeline-calibrated them (Table A1).

3. DATA AND METHODOLOGY

A carrier from an accelerating transmitter drifts in sky frequency, and over a long track its power spreads across many channels. The analysis therefore *de-drifts* the data, shifting each integration's spectrum to follow a trial drift rate before adding the integrations together.

3.1. *The de-drifted search statistic*

The search statistic is the largest stacked signal-to-noise ratio anywhere on the channel×drift grid of a window's dynamic spectrum,

$$T(\mathbf{x}) = \max_{\dot\nu,\,\nu} \frac{\sum_t I(\mathbf{x}, t, \nu + \dot\nu t) \,/\, \sigma^2(t, \nu + \dot\nu t)}{\left[\sum_t 1/\sigma^2(t, \nu + \dot\nu t)\right]^{1/2}}, \qquad (1)$$

where $I(\mathbf{x}, t, \nu)$ is the spectrum at sky position $\mathbf{x}$ in integration t and channel ν, $\dot\nu$ is the trial drift rate, and $\sigma(t, \nu)$ is the noise scale, both evaluated at the channel the track occupies in that integration. $T_\star$ denotes the value at the star. The time t runs from the first integration, so the frequency that labels a track is the one a carrier would occupy at the start of the block; a repeat epoch is read at the cell the same convention fixes (§4.6).

Equation 1 is the matched filter, and T an absolute signal-to-noise ratio, only if the noise is independent between integrations and channels, which the analysis does not assume. The trigger is instead set against the distribution T takes in the data, measured in every window at 512 control positions on the same visibilities and through the same stack. A value of T is converted into a power by injected carriers recovered through the unchanged pipeline (§4.1). The controls lie, from a fixed seed, in an annulus set by each block's synthesised and primary beams. The noise scale σ is their median absolute deviation at each integration and channel, scaled to a Gaussian-equivalent standard deviation, and the statistic is evaluated identically at the star and at each control. The field read over is the whole primary beam: every `FIELD` entry in which the star's primary-beam response reaches 0.05.

Spectra are extracted at the star's apparent position, never from images: for each integration and native channel the pipeline forms the weighted vector average of the phase-shifted visibilities in total intensity, $I = (XX + YY)/2$. That is the dirty-map value at the star, an amplitude and not a deconvolved flux density, so it is not a point-source measurement where the field holds extended emission. The correction for the star's offset from the phase centre is measured with injected carriers (Appendix A.2). Positions are propagated to the observation epoch for proper motion only; the annual parallax and orbital motion are corrected window by window in §4.1 and bounded for every component of a multiple in Appendix A.2. On-source time is counted per integration from the delivered exposure records. The data are the archive's deliveries, recalibrated with ALMA's final flags (and, for the six blocks calibrated by legacy scripts, its online flags and flux scaling) wherever a block sets a quoted limit or holds a crossing: 142 of the 178 such blocks, with 107 more that hold repeat epochs or runner-up windows. The other 36 keep the delivered calibration: 27 exceed the disk space available and in nine the recalibration did not finish in the time available; 25 of them set a quoted limit and 15 hold a crossing. On the 222 recalibrated Class A windows the rate at which control

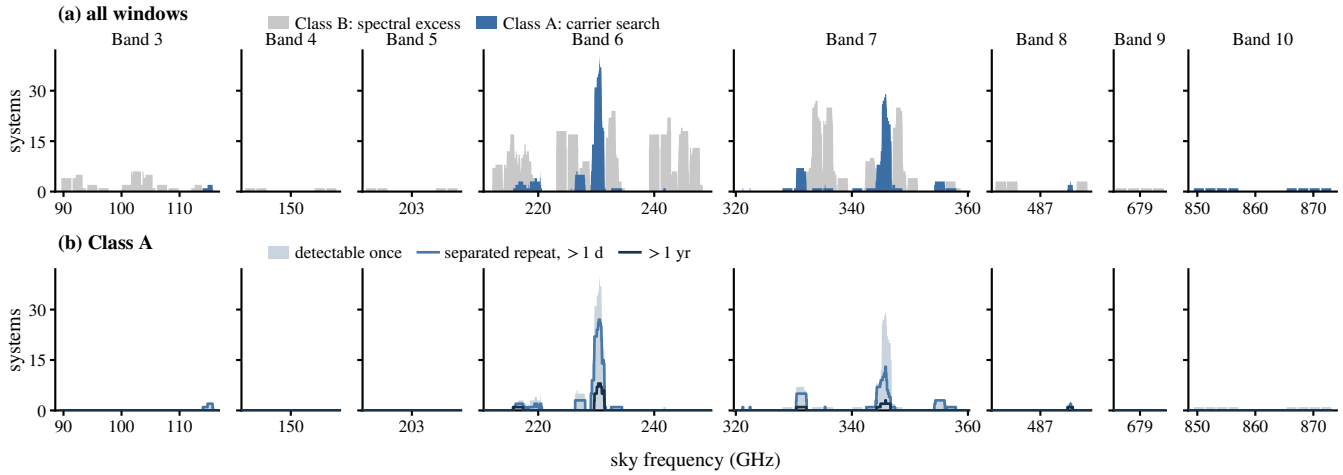


FIG. 2.— **Where this survey searched, and where a repeat could test it**: independent systems covering each sky frequency, one panel per populated ALMA band, its width proportional to the band's frequency share; curves step at the searched window edges. *(a)* Every window: blue, the fine-channel carrier search (Class A); grey, the coarse-channel spectral-excess search (Class B); the summed window bandwidth is 24.1× the union. *(b)* Class A: detectable once (grey); with a *separated* block (Table 2) covering the frequency (blue); beyond a year (dark). An accelerating carrier further needs a matching orbital phase.

positions exceed the trigger changes by a factor 0.999 (95 per cent interval 0.987–1.012), which bounds the chance expectation of the mixed calibration at 23.5–24.1. Every delivered channel is searched, without regridding, deconvolution or edge trimming. A per-integration median along frequency removes the continuum, and none is taken along time, so no feature narrow against the band can be removed. Because the delivered channels are smoothed, a correction computed from the correlator's own channel shape turns flux density times channel width into a radiated power (Appendix A).

The search runs on the topocentric sky frequencies ALMA delivered, and every drift rate quoted is the apparent one at the telescope. Two steps need the star's own frame: comparing a crossing with the line catalogue, and reading a repeat epoch at the *predicted cell*, the single channel and drift rate where the discovery's own fit puts a carrier. Both correct each block's frequency axis for the Earth's motion and then for the star's systemic velocity. That is the SIMBAD radial velocity (Wenger et al. 2000), queried on 2026 October 4, or zero where none exists – among the stars with a crossing, for ALMA J153702653−33192492 alone (§2). Toward β Pic the value of the ALMA CO analysis of its disc is adopted.[4]

Two power quantities are defined in Table 2. The trigger level $P_{\rm trig} = 4\pi d^2 S\,\Delta\nu$ is the power at distance d of a carrier that makes $T_\star$ reach $T_{\rm trig} = 5$ in one native channel of width $\Delta\nu$. The sensitivity EIRP_{90} is calibrated by injecting synthetic carriers of known amplitude into the calibrated visibilities of 49 windows and recovering them through the unmodified pipeline, one class factor being applied to every window. It is nominal in that it excludes atmospheric decorrelation, so it is not a fully calibrated completeness for a real carrier in every system (§4.1). Every limit quoted is the EIRP_{90} of its own window, beside its native channel width, except the labelled bare-threshold comparisons of §1; its error budget is Table 3. The amplitude scale is checked twice. An independent reduction of the same AU Mic visibilities is reproduced to 0.994 ± 0.014, which tests the extraction and not ALMA's flux transfer. The quiescent flux density of ϵ Eridani (Burton et al. 2022) is recovered to 1.10 ± 0.31, which measures the scale to 28 per cent. The absolute scale is therefore carried at ALMA's own accuracy specification: ±11 per cent for the sample and ±21 for the 14 Class A windows in Bands 8 and 10 (492–493 and 849–873 GHz).

The drift range is bounded by an acceleration, not by a frequency. Drift rate and line-of-sight acceleration are one quantity in two sets of units,

$$\frac{\dot\nu}{\nu} = \frac{a_{\rm los}}{c} \qquad \text{and so} \qquad a_{\rm los} = \frac{c\,\dot\nu}{\nu}, \tag{2}$$

so a ceiling on $\dot\nu/\nu$ is the same physical limit in every band. The default ceiling, 12.0–13.3 $\mathrm{Hz\,s^{-1}\,GHz^{-1}}$, is the acceleration of a transmitter orbiting a solar-type star at a few hundredths of an au. That is the Mason et al. (2025) fiducial with a threefold margin, and 6 per cent of the 200 nHz that Sheikh et al. (2019) recommend for a general search. It is widened for a planet host whose innermost known planet demands more: as an acceleration it is $3.60\,\mathrm{m\,s^{-2}}$, the median over the Class A windows, rising to $4.00\,\mathrm{m\,s^{-2}}$ on the 5 TRAPPIST–1 windows (Fig. 3). The grid is uniform in $\dot\nu$ and runs from the negative ceiling to the first trial at or beyond the positive one (Appendix D.5); it is fine enough that a carrier midway between two trials loses no more than 0.29 of a native channel over the longest track. **The survey does not constrain a transmitter whose apparent acceleration exceeds that range**: a carrier chirped faster (a chirp is not distinguished from acceleration), one not bound to a known planet, or one whose orbital and rotational accelerations together exceed it. The rotation of β Pictoris b, around a star searched here and the fastest measured for a planetary-mass object, alone corresponds to 19.4 nHz (Sheikh et al. 2019), above the ceiling. For transiting TRAPPIST-1 b, the most demanding known case, 29 per cent of the orbital phase lies above the default ceiling. The widened grid used on its star matches its maximum acceleration to within the ±3.1 per cent its

[4] $20\,\mathrm{km\,s^{-1}}$ (Matrà et al. 2017), which this survey's own blocks reproduce; it exceeds the catalogued value by about the template-mismatch offset the *Gaia* pipeline documents for stars this hot (Blomme et al. 2023).

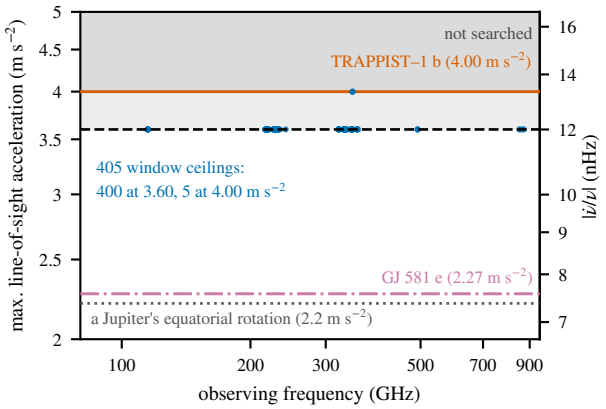


FIG. 3.— What the drift search reaches, as a physical selection function: the drift ceiling of each of the 405 Class A windows (points), by Equation 2, against observing frequency, with the unsearched region shaded and the same quantity as a fractional drift rate on the right. Horizontal lines are the reference accelerations of §3.1.

host's mass and orbit are known to, so between nothing and a sixth of the phase lies above that grid.

A drift search needs the maximum searched drift to cross of order one channel over the on-source track, $\eta_{\rm drift} = |\dot{\nu}_{\rm max}|\tau_{\rm track}/\Delta\nu_{\rm ch} \gtrsim 1$. 391 of the 405 Class A windows satisfy it; the other 14 fall below because their tracks are short, not because their channels are coarse. 38 of the 1258 coarse windows also satisfy it but stay in Class B, since crossing a channel is necessary for drift discrimination and not sufficient (Table 2); none of them holds a crossing. The per-window catalogue carries $\eta_{\rm drift}$ for every window.

3.2. *From a crossing to a confirmed signal*

Every window passes through three steps in the same order – the trigger, the line attribution and the recurrence test – which yield a *crossing*, an *unattributed crossing* and, had one survived, a *confirmed signal*, each defined in Table 2. None of the three is a significance test. Every count below is read from the ledger of every crossing the search produced, and is of primary crossings, one to a window.

The trigger. The level on $T_\star$ carries no tail probability. T is not shown to follow any standard distribution under the null, and a window holds between 112 and 6.87×10^6 channel×drift cells, so one level buys very different protection in different windows. Every cell that reaches it is recorded, not only the largest, because a carrier is masked as effectively by a noise excursion as by a line.

The background. A fixed threshold is not a fixed false-alarm rate, so the number of crossings expected by chance is computed window by window from the exceedance rate at that window's own control positions. That assumes a star behaves like a control position in its own field, which the reserved hold-out blocks test (§4.3.1). It also needs each block's control ensemble to behave like noise, and one block, A002_Xc7fa6f_X28a8 toward BD+05 1668, fails that criterion (Appendix D.2). With ALMA's final flags restored it holds no crossing, so the results are identical with it and without it. The census's quality-passing data hold 49 crossings, all of them Class A, and these are the baseline every expectation is set against. Of these, 1 lies in a window the same diagnostic flags inside a passing block; it is counted but withheld from the amplitude-dependent tests. The two in archival-tail blocks enter no expectation (Appendix D).

Two diagnostics say which crossing deserves attention first; neither decides a disposition or is charged to the sensitivity. The first ranks $T_\star$ against the window's 512 control positions, its lowest value, $1/(512+1)$, meaning that the star beat every control. It is a ranking filter and not a significance test, because star and controls are not demonstrably exchangeable. The annulus responds to whatever it surrounds, so genuine CO toward β Pictoris ranks below a control ring lifted by the resolved disc (Appendix B). Weighted star by star, the census median rank is 0.49 (0.47–0.52) and the reserved blocks' 0.46 (0.42–0.55), both consistent with the one half exchangeability implies. The second diagnostic, a visibility-domain fit, asks whether the emission is at the stellar coordinates, but at the trigger a noise maximum at the star also fits as a point source there (§4.5).

The line attribution is a single database query, made on 2026 October 6 and repeatable as it stands. A crossing within $\pm 50\,{\rm km\,s^{-1}}$ of a catalogued transition in the star's rest frame is *attributed*. One rule selects the species: covered by these setups and detected in discs or molecular clouds, whatever a window was tuned to (Appendix C). The half-width follows the Keplerian line-of-sight velocities of gas surviving beyond a few au, with margin for the systemic velocity. Halving it gives 21 attributed and 30 unattributed crossings at $\pm 25\,{\rm km\,s^{-1}}$, against 23 and 28 at $\pm 50\,{\rm km\,s^{-1}}$. A coincidence is a statement about a catalogue, not a property of a frequency, so an attribution is not a disposal: all 51 crossings go on to the recurrence test whichever side of the mask they fall. The price of the mask is that the survey constrains nothing inside it.

The recurrence test. A repeat counts only from a *separated* block, of another observing session and more than 1 day away, because blocks of one session share calibrators, atmosphere, correlator setup and interference. The day alone would not do: 31 pairs of blocks here are of one session yet more than a day apart, the widest by 12.91 d. Two readings are made. The *line-recurrence check* reads the one cell the discovery predicts in the star's frame, ± 0.08 to $\pm 0.63\,{\rm km\,s^{-1}}$ wide. It recovers an emitter at rest in that frame, as a line is, and a null there bounds only such an emitter, never a drifting or retuned carrier. The *trajectory comparison* asks whether one bounded trajectory inside the drift grid joins two crossings. It can reject an association between them but cannot accept one, and it cannot establish an artificial origin for either. Table 2 defines what a confirmation needs; only part of the sample has a separated block at all (§5.3).

The chain is shown to find what it is meant to find by the injected carriers of §4.1 and by β Pictoris CO, a real feature at a known stellar position, recovered and localised in two transitions and 10 execution blocks (§4.4).

4. RESULTS

The primary experiment is the Class A search for a drifting, unresolved carrier: 405 fine-channel windows toward 61 systems, 391 of them channelised finely enough to tell a drifting carrier from a stationary one (§3.1). The secondary Class B search, for an unresolved spectral excess with no drift discrimination, uses 1258 coarse

TABLE 2
The quantities and the three words this paper uses, defined once. Every term below is used in this sense throughout and in no other. The distinction the first three rows draw is the one the limits of §4.1 turn on: a crossing is a single-epoch event, and a confirmed signal needs a second and separated epoch that no sensitivity supplies.

Term	Definition
Crossing	A channel × drift cell at the stellar position reaching $T_\star \geq 5$ in one execution block; neighbouring channels above the trigger belong to one crossing. The largest in a window is its *primary* crossing and any other a *secondary* one, and because the largest cell is the window's own statistic a secondary crossing exists only where a primary one does. An event requiring investigation; no claimed significance.
Unattributed crossing	A crossing lying more than $50\,\mathrm{km\,s^{-1}}$ in the star's frame from every transition of the line catalogue. That is a reproducible frequency coincidence; which coincidences are astrophysically plausible is a separate judgement, stated with each one. A target for follow-up; not a detection.
Confirmed signal	The same carrier found again in a *separated* block – a different observing session, more than 1 day from the discovery – in one of two ways: recovered at the cell the discovery predicts in the star's frame (the *line-recurrence check*), or seen in three or more separated epochs that one orbit fits, the third where the other two predict it. The check recovers only an emitter at rest in the star's frame, as a line is, so a null there never excludes a drifting or re-tuned carrier. A pair of crossings that one bounded trajectory joins is *open*, not confirmed: the window searched, $\pm 801\,\mathrm{km\,s^{-1}}$, is wider than the $\pm 316\,\mathrm{km\,s^{-1}}$ an orbit at the drift ceiling can produce, so the joining test can reject a pair and cannot accept one (Table 5). For 21 Class A systems the archive holds no separated block at the frequency, so no sensitivity makes a confirmed signal possible there.
P_{trig}	The power of a carrier that makes $T_\star$ reach the trigger in one native channel. An internal pipeline level: not a sensitivity, not a significance.
EIRP_{90}	The power at which nine injected carriers in ten produce a crossing at the star: a measured single-epoch trigger completeness, not a detection probability. Calibrated by injection in 49 windows, one class factor applied to every window; each injected window's own point (×0.80–×1.58) is in the data archive. The factor is interpolated within the injected configurations for 294 windows and extrapolated beyond them for 62 (§4.1). Nominal in one respect: it excludes atmospheric decorrelation, which Table F1 gives system by system.
Class A	The primary experiment, defined by channel width: 405 fine-channel windows, median 488.3 kHz, toward 61 systems over 47.7 GHz of union bandwidth, 391 of which satisfy $\eta_{\mathrm{drift}} \gtrsim 1$ (§3.1).
Class B	The secondary experiment: 1258 coarse-channel windows, mode 15.625 MHz, constraining a persistent unresolved excess. Never a carrier search: in 1220 of them no searched drift leaves one channel, and in the other 38 the whole searched range moves a carrier 1.02–1.35 channels, which registers one but does not discriminate it (§3.1).
The sample	Neither volume-limited nor representative of the solar neighbourhood: the stars ALMA happened to observe for other reasons (§2).

windows. Together they are the 1663 windows of the primary census.

4.1. *Sensitivity: the power at which a carrier would have crossed the threshold*

The median system, on its best window, would have produced a threshold crossing from a carrier of 1.8×10^{15} *W.* Across the 60 systems with a constrained window the limits run from 8.7×10^{13} to 4.0×10^{16} W (Fig. 5); 24 of them reach 10^{15} W, and the median Class A *window* reaches 3.1×10^{15} W. Each limit is taken over the channels the line mask leaves in the star's own rest frame (§3.2). The finest windows were tuned to CO: a 15.26 kHz window spans 53.9 MHz against a mask 76.9 MHz wide, so 15 of the 405 Class A windows lie wholly inside it and constrain nothing, and HIP 10679 has no limit. Restricted to the 391 drift-resolving windows (44.4 GHz toward 58 systems), a system's limit is a median 14.1 per cent shallower. In 44 blocks whose in-beam fields were recombined after the limits were computed, the search ran on the recombined data, so the limits of the 8 systems such a block sets are conservative, by a median factor 2.1 and at most 5.7.

Carriers were injected at a random sub-channel phase and a random drift from each window's own grid, and recovered through the unchanged pipeline to the localisation at the star. Nine in ten come back at $3.03\,P_{\mathrm{trig}}$ in Class A, P_{trig} being the bare 5σ of the window itself, and that power is EIRP_{90}. It does not depend on the brightness of the control ring (Fig. 4), and the correlator's spectral response is carried by the injected tones (Appendix A). The 512-position spatial rank is not charged against it: that would cost a further factor 1.31, and in 3 bright-ring windows it could not be met at any injected power. Class B carries its own factor, $4.17\,P_{\mathrm{trig}}$, from 5 936 carriers injected into 35 coarse windows. The calibration interval of the class factor, ×0.89–×1.11 (Table 3), applies to the median and the counts and not to any one window. It places the median system at 1.6×10^{15} to 2.0×10^{15} W, so the second figure of that median is set by the calibration and not by the depth of any observation.

Carriers were injected directly into 49 of the 405 Class A windows, 12 per cent, and the result transferred to the other 356; that transfer is the dominant caveat on these limits. The injected windows, whole execution blocks drawn at random and balanced by band and array, match the population in the median to 6.3 per cent on five of 6 properties and differ by ×1.7 on the sixth, the noise level (Appendix F). Only the ninety per cent point in units of each window's own trigger power is carried across. Over the injected windows that point ranges from ×0.80 to ×1.58 of the adopted factor, 41 of the 49 lying within a tenth of it, and *that dispersion, not the calibration interval, is what one uninjected window's limit is uncertain by.* On the five properties that set the

reading, 294 of the 356 transferred windows lie inside the injected range on every one and 62 outside it on at least one, where that range is not an empirical bound. 8 systems take their headline limit from one of them, and none is among the 24 systems reaching 10^{15} W. The best window was itself injected into for 6 of the 60 systems, marked in Table F1; for the other 54 the headline limit is an estimate carrying that dispersion, so *the median is a calibration-transferred estimate and not a measured sensitivity*. Drawing each transferred window's factor from the 49 measured ones, once per execution block over 4 000 trials, lowers the median by 4 per cent and leaves 23–24 systems at 10^{15} W in nine trials in ten. Only TWA 3A, HD 285968 and HD 161868 can be moved across that level by the calibration interval, the spread or the decorrelation; each system's probability is in the data archive.

Every limit is nominal in one respect: it excludes atmospheric decorrelation, which an injected carrier does not suffer and a real one does. Estimated from each block's phase-calibrator phase rms (Cortes et al. 2026, §§10.4.11, 11.3.1) with the phase-scatter relation of Richards et al. (2022, Eq. 3), it leaves the median system's best window optimistic by 3 per cent; it exceeds a tenth for 12 systems and reaches 200 per cent for HD 172555. Divided out, it moves the median limit by 1.9 per cent and takes the 24 systems reaching 10^{15} W to 22. It is stated and not applied because only 279 of 407 blocks record the phase rms; the rest, estimated from blocks matched in baseline, are measurably the harder ones. Table F1 gives it system by system and the adjusted limits are in the data archive. The spectral response biases the other way. The injected tones fall 8 per cent short of the exact channel response of a window without online spectral averaging. For the 115 Class A windows delivered with averaging the exact factor is $1.83\,P_{\rm trig}$ (two-channel averaging, 114 of them) rather than $3.03\,P_{\rm trig}$, so their limits are conservative and would fall to $\times 0.60$ (Appendix A). That is stated and not applied, because no carrier was injected through the averaged response; an injection campaign using it is the remedy. *The net bias on one limit is therefore of either sign*: conservative by 8 per cent in the median Class A window, optimistic by 16 per cent at the ninetieth percentile of the systems' best windows.

A carrier that is not always radiating is penalised in proportion: in 3 888 undrifted trials down to a 10 per cent duty cycle the recovered amplitude followed it to 1 per cent, so a transmitter on a tenth of the time needs $10\times$ the power quoted; the drifting intermittent case is unmeasured.

4.2. *The nearest stars: Barnard's Star and Wolf 359*

Barnard's Star and Wolf 359 are the second- and third-nearest stellar systems if brown dwarfs are excluded, and to our knowledge these are the first millimetre or submillimetre technosignature limits for either; radio searches of the nearest stars have worked at centimetre wavelengths (Enriquez et al. 2017; Price et al. 2020; Margot et al. 2023). Both are non-detections, and *both limits come from the secondary Class B search*, the archive holding only coarse Band 6 windows toward them: Barnard's Star reaches 2.9×10^{14}–4.0×10^{14} W over 12 windows and Wolf 359 3.1×10^{14}–4.5×10^{14} W over 12.

4.3. *Threshold crossings and how each was disposed of*

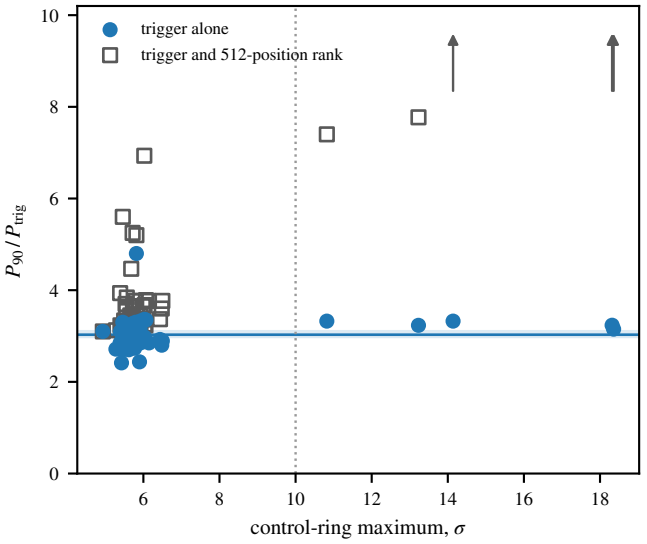


FIG. 4.— Why one completeness factor serves the whole class. The 49 directly injected Class A windows at the power recovering nine carriers in ten, each against its own 512-position ring maximum: no trend with the ring. Filled, the criterion adopted here, pooled value as line and band; open, the rank charged too, arrows where no power meets it; dotted, a resolved ring.

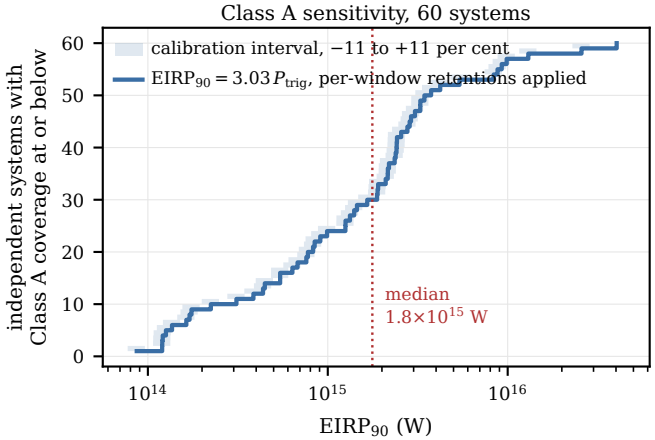


FIG. 5.— **Single-epoch trigger completeness of the carrier search**: the cumulative number of systems whose best window reaches a given $\rm EIRP_{90}$, each window's own retentions applied and atmospheric decorrelation excluded, the calibration interval of Table 3 shaded. Confirmation needs a separated repeat, which no power buys and which 21 of the 61 systems lack (Fig. 2b; §5.3).

Of the 1663 windows of the primary census, 49 hold a cell reaching $T \geq 5$ at the stellar position, all in data that pass the quality criterion of Appendix D.2; the window flag of that appendix reaches 5 windows in 2 blocks, and the 1 crossing it reaches, toward HD 48370, is withheld from the amplitude-dependent tests. The $\pm 50\,\rm km\,s^{-1}$ line mask places 23 of the 49 within that velocity of a catalogued transition in the star's own rest frame (Appendix C). Those 23 are attributed, 16 with corroborating evidence of line emission and 7 on the coincidence alone, and 26 are unattributed. The archival tail adds two more, both unattributed and both toward HD 14055, in windows no expectation covers (Table A1); they are in the ledger and in none of the counts above. The attributions are kinematic and not an accident of bandwidth: in the correlator's frame the crossings crowd onto the tran-

TABLE 3
What sets $EIRP_{90}$, and in which direction. The calibration interval applies to the class-average factor and so to any statistic over many windows; the spread between windows is an observed dispersion and is the statement about one window; the signed terms are biases and are not folded into an error bar (§4.1).

	value	what it does to the limit
Class-average completeness	×3.03	nine carriers in ten recovered at this multiple of a window's own trigger power. *Measured* in the 49 Class A windows injected directly and *estimated*, at this factor, in the other 356
Uncertainty on that calibration (per cent), combined in quadrature		
The factor itself	±2	sampling error of the pooled value over the 49 injected windows
Absolute flux scale	±10.2	the observatory's own band-dependent specification (ALMA Proposer's Guide (Privon et al. 2026), with the Technical Handbook's quality-assurance criterion (Cortes et al. 2026) at Band 8, where the two differ): ±5 in Bands 1–5; ±10 in Bands 6 and 7; ±20 in Bands 8–10, weighted over this sample's bands
Distance and visibility scale	±4.5	the Gaia parallaxes, which enter as d^2, and the block-to-block visibility scale
Combined	$-11/+11$	×0.89–×1.11 on the factor, and so on the median limit and on the system counts; ±21 for the 14 Class A windows in Bands 8 and 10, at 492–493 and 849–873 GHz
Spread between windows. An observed dispersion, not an error bar.		
Window to window	×0.80–×1.58	the 49 injected windows resolve 2.41–4.80 $P_{\rm trig}$ individually, 41 of them within a tenth of the pooled value and the second largest at ×1.11. This, and not the interval above, is what an estimated limit is uncertain by; it is wider, and it applies to 54 of the 60 systems
Applied window by window where a record allows it and bounded where none does, and so not combined with the above.		
Intra-integration smearing	×1.00–×1.05	a carrier sweeps through part of a channel within one integration and the injected ones do not: $\eta_{\rm smear}$ at the drift that sets a ninety per cent point, from each window's own integration length where it is recorded (1054 windows) and from the longest delivered anywhere where it is not (609)
Primary beam	×1.00–×4.3	the star is not at the pointing centre, so each limit is divided by the response at its own offset, on the full width the observatory documents. Nothing in the median window; the 12 windows furthest off-axis carry the departure of that form from a blocked circular aperture, up to 6.2 per cent, as a one-sided systematic
Annual parallax	×1.00–×2.06	the phase centre omits it, so each limit is divided by the amplitude a compact source retains at its own displacement, a median 0.13 per cent on the 1650 windows carrying a parallax record; on the other 13 the loss is bounded (Appendix A.2)
Signed terms. A bias is not an error bar and is not added to one.		
Channel response	×2.29	one over the share of a channel-confined carrier the peak channel keeps. Carried by the injections, so already inside the completeness above, which is this factor times a margin of ×1.325 over the nominal trigger
its residual	+8, +40	the injected profile falls short of the exact response, without and with online spectral averaging ($N = 1$, $N \geq 2$; Appendix A): *conservative*
Decorrelation	$+0/+200$	measured block by block from the phase rms the observatory records on each block's phase calibrator, which an injected carrier does not suffer and a real one does. One-sided, and a floor. A median +3 on the systems' best windows, +8 above 400 GHz, and more than +20 in 5 per cent of windows

sitions these setups were tuned to (Appendix E). Of the 49, two outrank all 512 control positions of their own window, 1 of them unattributed, and neither is present again at the cell its discovery predicts (§4.5). Behind each window's largest cell, twenty-eight weaker cells also reach the trigger; six are unattributed and none reaches the trigger in any covering block the recurrence test reads. No crossing survives the chain, and this survey reports no detection; the deposited crossing ledger lists every one.

4.3.1. *The unattributed count is what the control positions predict*

Because the searched channel × drift cells per window span 4.8 decades, the chance count is measured window by window over the 405 Class A windows of the census, with one condition on both sides of the comparison: a cell at the stellar position that reaches the trigger and that the line mask does not attribute (Appendix D). These fields' own control positions predict 23.8 unattributed Class A crossings and 26 are found. A count that large or larger arises by chance with probability 0.36, conditional on the control positions being a model of the stellar position, the one assumption the searched data cannot test. The expectation is not carried by a few noisy fields: without the ten windows whose control positions trigger most often, the other 395 predict 23.7 against 26 observed. In 12 of 12 subsets by band, array, channel width and field type the observed count lies within the central 95 per cent of its own predicted distribution. Calibrated instead at the stellar-to-control ratio 1.108 measured just below the trigger, the expectation is 26.4 at $P = 0.55$. A calibration error varying with frequency would raise the star and not its continuum-free controls, but the least demanding window would need 21.4 per cent of its star's continuum to reach the trigger. The excess below the trigger is not confined to windows with the star within one synthesised beam of the phase centre, where an additive visibility error would image (1.12 there, 1.07 beyond). It appears only in the 7 m-array data (1.15, against 1.04 on the 12 m array), and its cause is not identified (Appendix B).

Both sides are read over the whole primary beam, the field of §3.1. The narrower field is the extraction's original rule, which kept only the one entry whose phase centre lies nearest the star and so drops on-source time and real signal; neither rule has an adjustable radius. Read over that field on both sides, the same comparison gives 29 against 23.2 at $P = 0.15$. Neither geometry was nominated as the primary before the counts were seen; corrected for the 2 readings the smaller probability is 0.29, so neither departs from chance at the 5 per cent level, a correction that assumes only these 2 choices were made. Neither reading establishes a population of non-astrophysical transmitters. What the comparison could not have rejected is 12 further crossings against these control positions (16 on the narrower reading) – a count of crossings, not of transmitting systems, and not an occurrence rate (Appendix D).

4.3.2. *Stellar flares are excluded by a broadband prediction*

A millimetre flare raises every channel of a window alike. One able to produce a single-channel crossing must therefore produce a $5\sqrt{N}\sigma$ continuum event in the same integrations, 53–307σ for the windows holding a crossing; the observed continuum statistic has a median of −0.29 over the 48 measurable crossings, with $|Z| < 3$ in 45. One block tests the case directly: AU Mic `A002_Xa42f75_X319` holds both a crossing ($T_\star = 5.27$) and the millimetre flare of MacGregor et al. (2020), which raises the continuum at $Z = 10.3$ but contributes 0.17σ to the single-channel statistic, while the crossing's own de-drifted track reaches $Z_{\max} = 2.1$.

4.4. *β Pictoris: an astrophysical positive control, and the obstacle it identifies*

A search that finds nothing must show that it could have found something. The pipeline recovers the β Pictoris circumstellar carbon monoxide without being told the disc is there, in two rotational transitions and 10 execution blocks spanning 8.4 yr, drawn from 6 observing sessions. In the star's rest frame, at each block's own epoch, all 10 crossings lie within 0.5 km s^{-1} of the stellar systemic velocity, in the two lowest CO transitions of a star with mapped circumstellar CO (Dent et al. 2014; Matrà et al. 2017). It is also the positive control of the recurrence test: every feature recurs in another execution block, and each transition in another session. In one block, `X7116f1_X1785`, the resolved disc lifts the control ring to 14.11 against 9.68 at the star (Fig. B1): the rank tests compactness and not authenticity, and extended emission shows instead in the *level* of the ring, on which the window flag of Appendix D.2 is defined. HD 48370 is the complementary case, its crossing on the CO(2→1) that Cataldi et al. (2023) report there and flag as possible cloud contamination (Appendix C). The mask that removes such emission covers 12 species and costs 6.4 per cent of the Class A union bandwidth, and it is the single most productive test in the chain (§5.3).

4.5. *The two crossings that lead their own control fields*

Two crossings reach a stellar statistic above every one of the 512 control positions in their own window, and neither is a detection: 61 Vir at 346.3232 GHz and CP−72 2713 at 344.2965 GHz, both in Band 7 and in the star's own rest frame (Table 6; Fig. 6). Their rank says which crossing was examined first and disposes of nothing (§3.2). Were the stellar and control positions exchangeable, the control ensembles would predict 0.50 unattributed census crossings leading their field by chance, and one or more with probability 0.39. Each window contributes the share of its above-trigger controls that exceed every other position, weighted as in Appendix D. In the reserved blocks 0 of 5 crossings lead their field, and beyond 40 pc 0 of 6. CP−72 2713's crossing lies −12.3 km s^{-1} from SO(8_8–7_7) in the star's frame, inside the mask, so it is attributed on that coincidence alone (Appendix C). A visibility-domain fit, made for 39 of the 49 census crossings and consistent with a point source at the star for 33, rejects neither: injected sources in the bin these two occupy return $\mathrm{Re}/\sigma = 6.58 \pm 1.97$.

Neither crossing is present when the same stellar-frame frequency is observed again. The nearest covering block, 1.97 and 2.05 h from each discovery, holds a line-like emitter returning to the predicted cell below 0.95 and 0.71 of the discovery amplitude, and every covering block combined, over 3 and 2 d, below 0.55 and 0.65. Each carrier's own fitted drift would leave that half-channel cell (±0.211 and ±0.213 km s^{-1}) in 77 and 119 s, so this is a line-recurrence check and not a test of a drifting carrier (§4.6). At the faintest amplitude consistent with each crossing having been found, the check would have recovered such an emitter with calculated probability 1.00 and 0.66, and injected carriers measure 0.99 and 0.63, so for CP−72 2713 the silence does not exclude even a line-like emitter. In Table 6, T_{rep} is one cell at one trial and may be negative, and the recovery chance in brackets, at the discovery amplitude, is biased high. The readings behind each bound are in `candidates/repeat_epoch_readings.csv` of the data archive (White & Dey 2026).

4.6. *Recurrence, applied across the unattributed crossings*

The widened window. A transmitter in orbit changes its line-of-sight velocity between epochs, so the broadest reading asks whether anything reaches the trigger anywhere in a covering window, at any drift. That window is ±801 km s^{-1} wide in the median case, wider than the ±316 km s^{-1} an orbit at the grid's 3.6 m s^{-2} ceiling can produce about any star here. 14 of the 113 distinct covering windows hold a statistic above the trigger, against the 11.0 chance predicts ($P = 0.21$, or 10.0 at $P = 0.12$ when each window's own control positions supply the rate; Appendix D). Every one of those windows holds a crossing the ledger already lists, so the comparison re-reads the trigger-level count of §4.3.1 rather than adding a recurrence to it. 8 of the 14 are toward ALMA J153702653-33192492 and HD 14055 (§4.7); over the remaining 62 windows the reading is 6 against 4.5 at $P = 0.30$ (Appendix D.4).

The pair test. Table 5 asks of the 9 pairs of crossings inside that interval, across 4 stars, whether one transmitter could have produced both (Appendix D.4). Constant line-of-sight acceleration requires equal drift rates, and the fitted rates differ by more than the grid's resolution in 9 of the 9 pairs, so the question becomes a weaker one: whether any trajectory inside the searched grid joins the two. It separates 0 of the 9; of the 9 it cannot separate, 9 are *open* and 0 *edge-limited*, and an open pair is not a confirmation, since the comparison cannot accept an association (§3.2). At each census window's own control rate chance puts 3.8 open pairs in separated sessions (91–94 per cent of random pairs are joinable) against the 4 observed, pairs with an archival-tail member set aside.

The predicted cell. The *line-recurrence check* reads the one cell each discovery predicts at the repeat epoch, and so tests an emitter at rest in the star's frame, as a line is, and not a carrier. The 28 bounded crossings accelerate at 0.66 to 3.55 m s^{-2} (0.63 to 3.52 in the star's frame), which carries a carrier out of the half-channel in 38 to 471 s (median 105); each crossing's nearest covering block is at least 23 times its own residence away. Carrying the fitted drift forward does not rescue it: the drift's own resolution moves the predicted cell by a median 5 channels by the nearest repeat. On lines the check works: 13 of the 19 attributed crossings tested recur in another session, 39–1984 d away, and the 6 that do not are the faintest of them. On the 28 unattributed crossings whose cell could be read it returns nothing – at most

2.37 at the carried drift and 4.71 at any drift, against a trigger of 5. Over the 28 with a bound, a line-like emitter returning to its cell is held to a median 0.39 of the discovery amplitude and 0.997 at the weak end, where 5 are too shallow to test at the trigger amplitude. No bound depends on the cell's registration, the anchor or the threshold (Appendices B.2 and D.3). No crossing is confirmed in the sense of Table 2: none recurs at its predicted cell in a separated session, and three separated crossings that one trajectory joins pairwise occur toward one star, HD 14055, counting archival-tail blocks, and toward none in the census alone.

4.6.1. *The reserved hold-out and the control beyond 40 pc*

Three sets of execution blocks enter no census statistic (Table A1); each was searched with the same pipeline. The *hold-out*, 77 blocks, 315 windows, 36 stars and 238 h on source set aside by the identifier rule of Appendix B.1, holds 5 threshold crossings, the strongest at $T_\star = 8.64$, none outranking its own control positions. Of the 4 tested for recurrence exactly one is present again, a molecular line at $T_{\rm rep} = +8.47$; for the others the repeat coverage holds a line-like emitter in the predicted cell to a median 0.42 of the discovery amplitude and at most 1.06. The *archival extension*'s 463 windows in 115 blocks hold 10 crossings, 3 inside the mask in their own star's frame and 7 outside it, against 4.5 predicted by chance (Appendix D). The probability of 7 or more is 0.17, and the shortfall such a sample could have shown, $P(0\,|\,4.5) = 0.011$, is not seen. The *control beyond 40 pc*, 47 blocks and 198 windows toward 22 stars (90 h on source), holds 6 crossings, the strongest at $T_\star = 13.2$, none outranking its own controls. Read in the sky frame, it supports no limit and shows only that the pipeline behaves the same way beyond the sample. The hold-out crossings are released row by row with the crossing ledger.

4.7. *HD 14055 and ALMA J153702653−33192492*

Of the 14 covering windows above the trigger, eight are toward these two objects (Table 4 lists every crossing either holds). They are also the two most heavily observed objects in the survey, holding 20 per cent of the Class A census and 51 of the 113 covering windows, so 6.5 of the 11.0 expected exceedances are theirs before anything is looked at. HD 14055 holds 4 against 5.3 expected ($P = 0.81$; its own control positions predict 4.9): its concentration is its exposure. ALMA J153702653−33192492 holds 4 against 1.1 and 1.1 on the two nulls ($P = 0.024$ for this object alone). But it is the most discrepant of the 17 objects compared and was picked out for being so: chance puts some object that far out in 14 per cent of realisations. It is a target for follow-up, not evidence of a non-astrophysical source.

HD 14055's fields are ordinary: its control rings stand 1.04 times their class median, against 1.00 for the survey and 4.1 for the rejected block. Its setups are continuum ones, and its crossings lie 123–1069 km s^{-1} from the nearest masked transition in its own frame. The 651 d baselines of Table 5 come from its two archival-tail crossings. The 9 epochs carrying the inverse variance of those readings are follow-up blocks outside the census, and on them an emitter at rest in the stellar frame at the discovery amplitude (a line-like source) is excluded at 11.8 and 12.7σ; this does not test a drifting carrier (§4.6).

TABLE 4
Every crossing toward HD 14055 and ALMA J153702653−33192492. Rates are in the observatory frame; ≥ and ≤ mark a fit on the grid's outermost trial.

$\nu_\star$ (GHz)	$T_\star$	$\dot\nu_{\rm obs}$ (Hz s^{-1})	nearest masked transition	$\Delta v_\star$ (km s^{-1})	Re/σ
HD 14055					
331.254	5.10	+1635	^{13}CO(3–2)	+604	+5.1
331.766	5.57	−3857	^{13}CO(3–2)	+1069	+5.0
344.452	5.01	−2589	SO(8_8–7_7)	+123	–
344.696	5.18	−2426	SO(8_8–7_7)	+336	+4.8
345.092	5.17	+2123	SO(2_3–2_1)	−531	+4.6
345.430	5.45	+2584	SO(2_3–2_1)	−238	–
ALMA J153702653−33192492					
216.204	5.25	−2198	SiO(5–4)	−1245	−0.6
217.727	5.09	−983	$H_2CO(3_{0,3}$–$2_{0,2})$	−680	+1.8
218.669	5.05	−2023	$H_2CO(3_{2,1}$–$2_{2,0})$	−125	+4.5
218.892	5.11	+2373	$H_2CO(3_{2,1}$–$2_{2,0})$	+181	+4.7
230.539	5.39	−19	CO(2–1)	+2	+4.4
347.189	5.01	−2581	SiO(8–7)	−122	+4.6

TABLE 5
Can two crossings be one carrier? The 9 pairs toward one star with an unattributed member (§4.6); the note below defines the columns and the verdicts.

Star	blocks	new sess.	Δt (d)	$\dot\nu_{1\star}$ (Hz s^{-1})	$\dot\nu_{2\star}$	$\Delta\nu$ (MHz)	miss grid	one?
61 Vir	X822/X82f	yes	3.00	+1092	+3221	+755.4	15	open
J1537−3319	X10359/X398	yes	123	−2000	+2390	+223.5	3.6	open
HD 14055	X24be0/Xd46	yes	1.92	+1665	−3823	+512.7	100	open
	X2b2ee/X3835	yes	2.09	−2390	+2155	+395.5	55	open
	X3835/Xd526	yes	607	+2155	−2563	−640.4	–	open
	X2b2ee/Xd526	yes	609	−2390	−2563	−244.9	–	open
	X3835/X25c5e	yes	649	+2155	+2611	+338.3	–	open
	X2b2ee/X25c5e	yes	651	−2390	+2611	+733.8	–	open
β Pic	X3a90/Xa9df	no	1.02	−1611	−679	+85.5	132	open
HD 285968	X108f/X5b4e	yes	4.99	−29	−29	−0.002	0.6	yes
	X3bf6/X108f	yes	39	−29	−29	−0.024	0.6	yes
	X3bf6/X5b4e	yes	44	−29	−29	−0.027	0.6	yes

New sess.: *yes* where the two blocks belong to different observing sessions. Starred rates are in the star's frame, the observatory's own acceleration removed (observatory frame: Appendix D.5). *Miss* measures departure from constant acceleration, in units of the drift grid's resolution over Δt; a dash marks a pair that resolution cannot separate. The verdict is set by the bounded-trajectory test, which a constant-acceleration failure does not decide – a miss of 3 or less does, and sets *yes*; *edge*, one rate lies on the grid's outermost trial and is a lower bound. Below the rule, a control: the CO(2–1) line of HD 285968, which does not drift.

ALMA J153702653−33192492 is a serendipitous dwarf 5.75 arcsec off the phase centre of fields pointed at the Lupus clouds, in 212 s snapshots. Its CO(2 → 1) crossing is cloud emission filling the field: it sits on a line the observers tuned to, with a fitted drift 0.6 of a grid step from zero and a control ring reaching $T_\star = 13.4$ against 5.4 at the star. The star has no measured systemic velocity, and recomputing the attribution over ±50 km s^{-1} changes only that crossing, whose reading in the local standard of rest holds anyway (Appendix C). What is left is four windows where 1.1 were expected, on four crossings between 216 and 220 GHz in two of this object's eight blocks. Their control rings are ordinary, their drifts far from zero, and a visibility fit places two of the four at the star. A transition of a wider 225-species list lies within 15 km s^{-1} of every crossing in Table 4, as its 24 976 transitions guarantee, so a wider list identifies nothing. We have no explanation. A repeat observation should point at ALMA J153702653−33192492, with a radial velocity and one deeper epoch at 216–220 GHz.

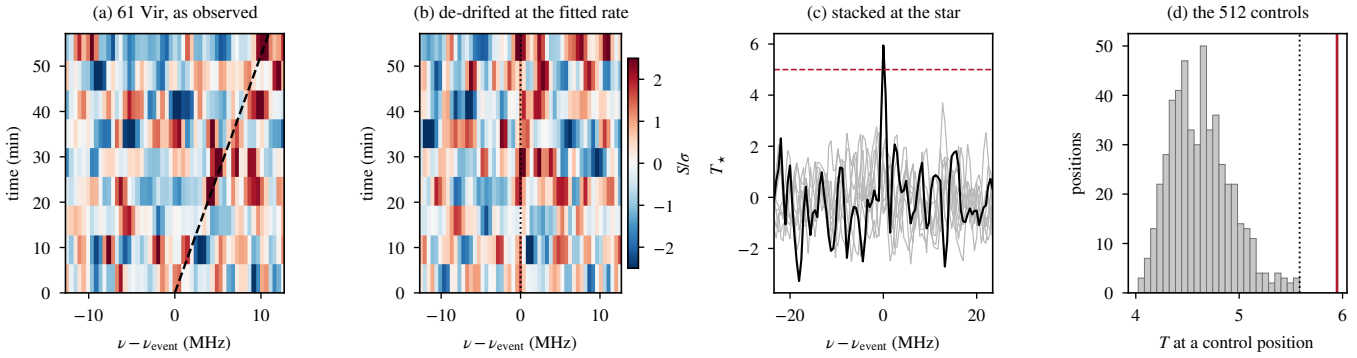


FIG. 6.— **The strongest unattributed crossing of the quality-passing data, and the control field it leads.** Frequencies are in the star's rest frame. *(a)* The dynamic spectrum at 61 Vir's position around the crossing at 346.3232 GHz, observed 2017 May 20, the fitted drift of $+3182\,\mathrm{Hz\,s^{-1}}$ dashed, 225 integrations binned into 9 and plotted in units of each bin's noise. *(b)* The same rows shifted by that drift. *(c)* The de-drifted stack at the star (black) and at the 8 control positions with retained dynamic spectra (grey), in units of its own noise, the trigger dashed. *(d)* The same statistic at each of the 512 rank-screen control positions: the star (solid) against the highest control (dotted).

TABLE 6
THE TWO EVENTS, THE CROSSINGS WHOSE STELLAR STATISTIC EXCEEDS ALL 512 CONTROL POSITIONS OF THEIR OWN WINDOW. §4.5 READS THEM.

	61 Vir	CP−72 2713
Frequency, stellar frame (GHz)	346.323236	344.296455
Drift rate ($\mathrm{Hz\,s^{-1}}$)	+3182	−2056
$T_\star$	5.947	5.809
Highest control T	5.58	5.68
Fitted amplitude (mJy)	118.6 ± 19.9	11.3 ± 2.0
Peak-channel EIRP (W)	5.0×10^{14}	8.9×10^{14}
Repeat epochs measured	3	5
Nearest covering epoch, separation (h)	−1.97	+2.05
T expected if persistent, $T_{\rm pers}$	9.69	6.29
T measured, matched drift, $T_{\rm rep}$	+0.36	−0.88
Amplitude still bounded, $(T_{\rm rep}+5)/T_{\rm pers}$	0.55 S	0.65 S
Calculated recovery chance Φ at $5S/T_\star$ (at S)	1.00 (1.00)	0.66 (0.90)
Nearest masked transition	$\mathrm{SO}(9_8\text{–}8_7)$	$\mathrm{SO}(8_8\text{–}7_7)$
its offset $\Delta v_\star$ ($\mathrm{km\,s^{-1}}$)	−177.6	−12.3
Disposition	not confirmed	attributed

4.8. *A stellar-frame multi-epoch stacked search*

Combining a star's epochs in its own rest frame deepens the limit. 59 of the surveyed stars can be stacked, 5 of them from blocks beyond the census. Over 316 stacked groups, 2191 epoch-windows and 1429 h the flux-density limit improves on each star's best single epoch by a median factor 1.50 and by 4.45 at most. A further 10 stars, all control stars beyond 40 pc, are stacked and excluded from those figures. *A fixed stellar-frame frequency is not a fixed observed frequency*: the barycentric term alone swings by a median $8.0\,\mathrm{km\,s^{-1}}$ between the epochs of one star, many channels at the 488 kHz Class A width. Each epoch is therefore transformed to the stellar rest frame, divided by its own primary-beam response and combined by inverse variance over the channels of common stellar-frame coverage. The transform is verified on HD 285968's line, which 4 epochs observe $12.2\,\mathrm{km\,s^{-1}}$ apart and place at one stellar-frame frequency to $0.03\,\mathrm{km\,s^{-1}}$. Every stacked limit is measured by injection into the stack, and the criterion is applied identically at the stacked control positions. *The gain is* $\sqrt{N_{\rm eff}}$ *and not* $\sqrt{N}$, where $N_{\rm eff} = \sigma_{\rm min}^2 \sum_e \sigma_e^{-2}$ is the equivalent number of equal-depth epochs: the median $N_{\rm eff}/N$ is 0.65 and the realised gain against $\sqrt{N}$ is 0.79, the shortfall being unequal epoch depth.

The deepest stacked limit is Proxima Centauri's, and it is a Class B limit: its 25 census blocks are all coarse 15.625 MHz windows, so the stack constrains an unresolved spectral excess, discriminates no drift and reaches $\mathrm{EIRP}_{90} = 2.2\times10^{13}$ W. *Every stacked limit holds only for an emitter steady in the stellar frame to within one of its own channels*, and a transmitter on Proxima Centauri b is not. Its orbit (Anglada-Escudé et al. 2016; Kervella et al. 2017; Akeson et al. 2013) swings it over $94\sin i\,\mathrm{km\,s^{-1}}$, some 5 channels at $\sin i = 1$, which the stack would cancel unless the orbit is seen nearly face-on ($\sin i \leq 0.22$, 2.5 per cent of random orientations). Within one block it moves at most 6 per cent of a channel, so the single-epoch limit of 5.51×10^{13} W, from a reserved block and not a census limit, is the one that applies to it. No crossing of §4.6 drifts slowly enough to survive a fine-channel stack, which admits far less stellar-frame excursion than a coarse one, and the single-epoch limits of §4.1 are the ones that apply generally. *2 groups pass the full criterion and both are known lines*: β Pictoris carbon monoxide in two transitions, the positive control of §4.4 passing in the stack as well. The survey's own $\pm50\,\mathrm{km\,s^{-1}}$ mask leaves none of them unattributed.

5. DISCUSSION

5.1. *What was searched*

The search space opened here is millimetre and submillimetre frequency toward the nearest stars: 47.7 GHz between 114 and 873 GHz toward systems within 40 pc, where the only other search of more than a single line covered 117 MHz of Band 3 (§1). The channels were inherited, not chosen: from 15.3 kHz to 31.25 MHz, against the 2.8 Hz of centimetre-band carrier searches (Fig. 1(b)).

Union bandwidth overstates what that buys, because frequencies were searched toward different stars to very different limits. Figure 7 therefore counts the systems in which a carrier of a given EIRP and sky frequency would have been recovered nine times in ten. At 1×10^{15} W the covered frequency falls to 19.0 GHz, 40 per cent of the union, and reaches 24 systems, 16 of them near 230 GHz. At any power that frequency reaches 38 of the 61 systems, and half the sample is reached only above 4.3×10^{15} W.

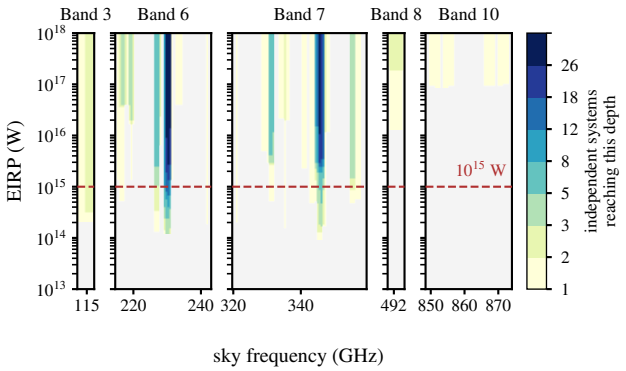


FIG. 7.— Where the experiment could have found something. Shading, the independent systems recovering an unresolved carrier of the plotted EIRP and sky frequency nine times in ten *in a single epoch*; unshaded, none. Confirmation needs a second epoch covering the same frequency, which no power supplies. Panels as in Fig. 2b, which gives that coverage. Dashed, 1×10^{15} W.

5.2. *What the non-detection constrains*

Every limit here applies to one kind of signal: an unresolved carrier at the stellar position, confined to a single 15.3–1953 kHz channel, lying outside the molecular-line mask, radiating while an archival block was on sky, and drifting linearly within its window's ceiling of 12.0–13.3 nHz. That covers every known planet of the sample except TRAPPIST–1 b, reached only as precisely as its host's mass and orbit are known (§3.1); a transmitter on a rapidly rotating planet spends most of it on spin (Fig. 3). Emission broader than a channel needs close to 32× the power for a feature as wide as the widest line in these fields, and a carrier on for part of a block is recovered at that fraction of its amplitude. Nothing is constrained inside the $\pm 50\,\mathrm{km\,s^{-1}}$ mask, or toward the stars within 40 pc that ALMA has not observed, almost all of the 17 566 this volume holds. The result is therefore a non-detection, not an exclusion of persistent transmitters, for the reasons of §5.3.

Two blind spots. The analysis forms (XX+YY)/2, which retains the full intensity of a circularly polarised carrier. The loss is not sensitivity but the discrimination polarisation would offer against unpolarised astrophysical emission: ALMA's feeds are linear, and the circularity lies in the cross-hands, which this analysis does not extract. And the 203 GHz positronium hyperfine line that Steffes & DeBoer (1994) and Mauersberger et al. (1996) observed lies in ALMA Band 5, where this survey holds 4 coarse-channel windows, all toward α CMa B, and none of them covers it.

5.3. *What prevents stronger conclusions*

Depth is not the main limit. An illustrative benchmark, a 12 m dish radiating 1 MW at 230 GHz with an aperture efficiency of 0.7, has a directive gain $G = \eta(\pi D\nu/c)^2 = 5.9 \times 10^8$ and so an EIRP of 5.9×10^{14} W, above the single-epoch limit of 16 of the 60 Class A systems that retain unmasked bandwidth. Its beam, $\Omega = 4\pi/G = 2.1 \times 10^{-8}$ sr, illuminates 1.7×10^{-9} of the sky, so EIRP is not transmitter power: these limits apply to a beamed transmitter only if its beam contains the Solar System. What limits the archive is the astrophysics that imitates a carrier, and the repeat coverage it happens to hold.

Line emission, not interference, is the false positive at these wavelengths. Catalogued molecular transitions account for 23 of the 49 crossings in the census's quality-passing data, and interference for none (Appendix E), leaving 26 for the comparison with chance of §4.3.1. Of the attributions, 16 have support beyond the frequency match (a second transition of the species, catalogued gas on the sightline, or the crossing's own recurrence) and seven rest on the coincidence alone. A line filling the primary beam raises the star and its control annulus together, a discriminant with no centimetre analogue: 10 of the 12 windows that reached the spatial stage hold disc or interstellar carbon monoxide (§4.4).

Repeat epochs turn a crossing into a testable claim, and an archive supplies them only by accident. Let the *confirmation coverage* $C_{\mathrm{conf}}(\nu)$ be the number of systems toward which two execution blocks of different observing sessions, more than a day apart, both cover the sky frequency ν; it is stated against frequency because repeat visits re-tune. Its support is 24.0 GHz, 50 per cent of the 47.7 GHz searched, and 8.5 GHz beyond a year (Fig. 2b). Of the 61 systems, 40 have a second session more than a day away covering a frequency of the first and 13 more than a year away; for the other 21 no power could have made a confirmation possible. Where a repeat exists it can be too shallow to test anything, as for 5 of the 28 crossings read at a predicted cell, and the line-recurrence check it supports cannot test a carrier that drifts or is re-tuned between epochs (§4.6).

5.4. *What a dedicated ALMA experiment should do differently*

Each limit has a remedy on the same telescope. *Channelise finely everywhere*: the 1258 coarse-channel windows cannot carry a carrier search. *Choose the targets and the frequencies*: this archive favours dusty discs, the worst case for line confusion, and covers least the M dwarfs that host most nearby habitable-zone planets; windows placed away from the bright transitions would recover most of the 3.05 GHz of Class A union that the mask costs. *Repeat the epochs on purpose, at the same tuning*, days and years apart: the separation is a scheduling constraint rather than extra sensitivity, and three separated epochs over-determine an orbit that two can only fit. 27 of the 61 systems already carry three at a common frequency, so the shortfall is a candidate seen at more than one of them. *Retain the visibilities*: with the phase centre at the star, a genuine point source lies entirely in the real part of the visibility, which asks a stronger question than a rank against neighbouring positions. The first targets are HD 14055 and ALMA J153702653−33192492 (§4.7).

6. CONCLUSIONS

We searched archival ALMA observations of 61 star systems within 40 pc for narrowband, drifting radio carriers. The fine-channel spectral windows searched cover 47.7 GHz of sky frequency in 26 intervals between 114 and 873 GHz, at least 1363 times the bandwidth of either millimetre search of nearby stars before it (Steffes & DeBoer 1994; Mauersberger et al. 1996).

Injection of synthetic carriers into 49 windows, transferred to the rest, gives a median of 1.8×10^{15} W of equivalent isotropic radiated power for a carrier that a

system's best window registers nine times in ten. Of the 60 systems with a constrained window, 24 reach 10^{15} W, or 22 allowing for atmospheric decorrelation.

The census holds 49 threshold crossings in data that pass the quality criterion, and 23 of them are attributed to catalogued molecular lines, mostly carbon monoxide. The other 26 are consistent with chance: control positions beside each star predict 23.8 ($P = 0.36$). Where a crossing's frequency was observed again, 14 of the 113 repeat windows reach the threshold against 11.0 expected ($P = 0.21$), eight of them toward two objects. HD 14055's share is explained by its exposure; ALMA J153702653−33192492 is a target for follow-up, not an established excess, since chance puts the most discrepant of 17 objects that far out in 14 per cent of trials.

No transmitter is detected: of the unattributed crossings whose repeat could be read, none returns at the frequency its discovery predicts. This is a non-detection, not an exclusion. The limits apply only to a carrier outside the molecular-line mask, on while ALMA observed and drifting by at most 12.0–13.3 nHz (3.60–4.00 m s^{-2}; §5.2). Confirmation, more than sensitivity, limits what the archive can show: 21 of the 61 systems were never re-observed at a covered frequency, and the line-recurrence check can recover an emitter at rest in the star's frame but not a carrier that drifts or is re-tuned between epochs. A dedicated programme would channelise finely, avoid the bright molecular lines and revisit each target at one tuning, days and years apart (§5.4).

ACKNOWLEDGEMENTS

This paper makes use of the following ALMA data: the 67 projects listed in Table A3, each cited as `ADS/JAO.ALMA#` followed by its project code. The 407 primary-census execution blocks searched span 2013 October 6 to 2025 July 11, and we thank the principal investigators and teams who proposed and obtained them. ALMA is a partnership of ESO (representing its member states), NSF (USA) and NINS (Japan), together with NRC (Canada), NSTC and ASIAA (Taiwan), and KASI (Republic of Korea), in cooperation with the Republic of Chile. The Joint ALMA Observatory is operated by ESO, AUI/NRAO and NAOJ. This work has made use of data from the European Space Agency (ESA) mission *Gaia* (Gaia Collaboration et al. 2023), processed by the *Gaia* Data Processing and Analysis Consortium (DPAC). This research has made use of the SIMBAD database (Wenger et al. 2000) (CDS, Strasbourg, France); of the NASA Exoplanet Archive (Akeson et al. 2013), which is operated by the California Institute of Technology under contract with the National Aeronautics and Space Administration under the Exoplanet Exploration Program; and of the Splatalogue spectral-line database (Remijan et al. 2007).

Use of generative artificial intelligence. Large language models were used in this work: they wrote most of the software – the archival retrieval and calibration scripts, the search and vetting pipeline, the statistical tests, and the generators that produce every number, table and figure in this paper. They also drafted text, which the authors revised. The authors set the scientific question, chose every criterion, and are responsible for the content of this paper.

Testing and validation. No result here is accepted because a model reported it: every number in the text is written by the program that computes it from the data, and a separate audit re-derives the identities that must hold among the published values and refuses the paper where one does not. Every check in the analysis was first made to fail on deliberately wrong input before it was relied on.

Software. The pipeline depends on CASA 6.7.0 (CASA Team 2022), PYTHON 3 with ASTROPY (Astropy Collaboration et al. 2022), NUMPY (Harris et al. 2020) and SCIPY (Virtanen et al. 2020), and the ALMA Science Archive and SIMBAD TAP interfaces through ASTROQUERY (Ginsburg et al. 2019); all are free and open-source, at the versions cited.

Author contributions. G.J.W. conceived the survey, defined the target sample and the selection criteria, and wrote the manuscript; R.D. built and operated the archival retrieval, calibration and search pipeline, implemented the spatial control statistic and the injection–recovery campaigns, and produced the per-window catalogue and its regeneration scripts. Both authors designed the vetting procedure, chose the masked species, carried out the statistical validation reported in §3.1 and Appendix D, and agreed the dispositions of the crossing ledger.

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Competing interests. The authors declare no competing interests. VBRL Holdings is a privately held company through which R.D. conducts independent research, named here solely as his affiliation; it had no role in the design of the survey, the selection of targets, the analysis, the decision to publish or the content of this paper, and provided no funding for the work.

DATA AVAILABILITY

Every observation used here is public in the ALMA Science Archive under the 67 project codes of Table A3, and is retrieved by the execution-block identifier, which the crossing ledger gives for each of the 51 threshold crossings (Chen et al. 2022); the astrometry is Gaia DR3, the planet counts the NASA Exoplanet Archive, and the line mask one Splatalogue query of CDMS and JPL made on 2026-10-06. The machine-readable data and the software are archived on Zenodo (White & Dey 2026). The principal results are reproduced from the per-window catalogue of all 1663 windows, the Class A and Class B sensitivity measurements, the crossing ledger, the control-field statistics and the execution-block accounting (every row of Table A2 and Table A1): from these tables the counts, the chance expectation of 23.8 crossings, the EIRP_{90} summaries and the line-recurrence bounds rebuild row by row, and the archived generators reproduce every number, table and figure printed here; the principal tables also accompany the arXiv version of this paper as ancillary files. The archive also holds the search pipeline as it ran, its known defects recorded rather than removed, short reference implementations of Equation 1, the control positions, the injection harness and the line mask, and instructions for taking one public execution block to a statistic printed here. The dynamic spectra at the star and at its 8 retained control positions, around each of the 51 crossings, form a second file of the same

record (169 MB). Not public are the calibrated visibilities, of order a terabyte for the 407 blocks, and the per-window spectra, which were discarded as the search consumed them for lack of storage and not to restrict access; this execution of the search therefore cannot be repeated, only run again from the archive with the same code. For the same reason two quantities do not rebuild from the tables alone: the interval on the chance expectation, whose resampling draws on quantities that are not per-window, and the trials behind the 49 injected windows' ninety-per-cent points, although the points themselves are archived. Windows whose blocks were not recalibrated carry products extracted before the archived extractor was finalised; rerunning ten census blocks drawn at random among those under 1.8 GB (40 windows) with it moves $T_\star$ by a median of 0.02, by 0.1 or more in two windows and by at most 0.24, and leaves their noise unchanged to 0.1 per cent.

APPENDIX

A. SPECTRAL RESPONSE, FREQUENCY CONVENTIONS AND THE PRIMARY-BEAM RESPONSE

A.1. *Frequency frames, linewidth and smearing*

Every frequency axis here is *topocentric*, as the correlator delivers it; velocities enter only at vetting, where the line mask carries each crossing to the stellar frame (Appendix C). $\mathrm{EIRP}_{\mathrm{trig}}$ is the minimum *total* power of a carrier confined to one native channel, never a spectral power density, as are the Hz-resolution surveys' limits in Fig. 1(a). Spreading it over N channels lowers the statistic as $N^{0.81}$, so a feature as wide as the widest line these fields resolve (69 channels) must carry 32× a carrier's power.

ALMA's online Hanning smoothing (Cortes et al. 2026) leaves the peak channel 0.50 of an unresolved carrier's flux at a channel centre and 0.42 on a boundary. *That loss enters the limits exactly once.* Each injected carrier is deposited with the correlator's channel response at a sub-channel phase drawn afresh and recovered through the unmodified trigger, so the measured completeness contains the loss and factorises into it: $3.03\,P_{\mathrm{trig}}$ is ×2.29 for the channel response and ×1.325 for the margin a peak channel needs above the nominal trigger for nine carriers in ten to return. P_{trig}, S_{min} and the filled symbols of Fig. 1(a) are therefore nominal and comparable with other programmes' thresholds, while EIRP_{90} carries the correction.

The factor $2.29 = 1/0.4375$ is set by the phase-mean peak-channel share of the 0.25/0.5/0.25 kernel, which the injections deposited in every window. For the 290 Class A windows delivered without online spectral averaging ($N = 1$) that share is 8 per cent short of the exact 0.4741; for the other 115 ($N \geq 2$), whose broader response the injections did not use, it is 40 per cent short of 0.726. The completeness is conservative by that much: for $N = 2$ the exact factor is $1.83\,P_{\mathrm{trig}}$, which a refit of the campaign's own tones reproduces. The limits are not corrected for it (§4.1); each system's limit with the exact response is in `sensitivity/system_limits.csv` of the data archive.

Intra-integration smearing, which the injections do not model, is divided out of every limit once as η_{smear}, the mean channel response over the sweep within one integration. It exceeds one per cent in 9 windows, reaching ×1.05; it is measured for 1054 of the 1663 windows and bounded for the other 609 at 11.6 s, the longest integration delivered. A re-scoring of the injected windows shows that this correction errs the safe way (`documentation/appendix_A_smearing.md` of the data archive, White & Dey 2026).

A.2. *The primary-beam response*

The sensitivity quoted for a window is $S_{\mathrm{min}} = 5\sigma/A$, where A is the primary-beam response at the star's offset from the phase centre. Archival fields are pointed at catalogue J2000 positions, so these high-proper-motion stars can sit off axis: a median 0.089″ and at most 17.99″ over the 1663 windows, with A falling to 0.2308 and below 0.9 in 46 of them. *The response applied is a Gaussian of the documented full width* $1.13\,\lambda/D$ (Cortes et al. 2026) at each window's own dish diameter. It matches the blocked-aperture response of CASA (CASA Team 2022) to 2 per cent in width; the two part company with offset, which sets the retention rule of §2, and *at the furthest census window,* $0.82\,\lambda/D$*, they agree to 6.2 per cent, the largest beam-model uncertainty any sensitivity here carries.* The correction is applied once, to S_{min}; the statistic, ranks and attribution are ratios within one window, in which A divides out.

The annual parallax is measured rather than modelled: carriers injected at the full apparent position of the 8 furthest-out windows retain 0.92 and 0.58 of their amplitude where the model predicts 0.89 and 0.49, so a model-corrected limit is shallower than it need be. ϵ Eridani, with 24 of its 44 windows 0.75–0.83 λ/D off axis and the largest parallax in the search, has no parallax record, so its limit is optimistic by at most 9 per cent.

The propagation assumes straight-line motion, which a component of a bound pair does not follow. 30 of the 91 searched positions (the 90 census stars and HD 139084 B) have a companion within 400 arcsec (Mason et al. 2001; Matson et al. 2026); the displacement, from a visual orbit, the separation with masses from Pecaut & Mamajek (2013), or the proper-motion anomaly (Kervella et al. 2022), moves no system's limit by more than 16 per cent. Barta 161 12, HD 155555C and LHS 1140 have disturbed solutions with no companion and no Hipparcos epoch, so the 24 windows of those 3 stars carry limits optimistic by an unmeasured amount.

Table A1 accounts for every execution block the query exposed and Table A2 for every event; their itemisation, all 29 rows under the same row numbers, is `catalogues/count_accounting.csv` of the data archive. The census's on-source time is 281 h; planet counts are from the NASA Exoplanet Archive `pscomppars` table (Akeson et al. 2013), accessed 2026 October 4.

B. THE SPATIAL RANK, AND WHAT IT CAN SUPPORT

The spatial rank asks whether an excess sits at the star or anywhere else in the field. It is a diagnostic on which no disposition rests (§3.2). The star lies near the phase centre, inside the control annulus, and carries more above-threshold cells than a control position at every level below the trigger (Appendix D). *This is not a primary-beam effect*: spectra are ex-

TABLE A1

The survey ledger, part one: execution blocks and windows. Each row carries its own number, and a group sums to the row whose number heads it. Hours are window-hours, so the classes partition the hours as they partition the windows. Rows 3.2 to 3.5 are the archival gap, over which no result is formed.

	Set of execution blocks	Blocks	Windows	Stars	Systems	Hours
1	Archival scope: blocks the query exposes	656	–	–	–	–
1.1	… processed with the same pipeline	479	–	–	–	–
1.2	… not processed at the freeze	177	–	–	–	–
2	Processed with the same pipeline[c]	484	–	–	–	–
2.1	**Primary census**	407	1663	90	83	1098
2.1.1	Class A, fine channels[a]	–	405	66	61	228
2.1.2	Class B, coarse channels[a]	–	1258	89	82	870
2.2	Reserved hold-out	77	315	36	–	238
3	Not processed at the freeze: what became of each	177	–	–	–	–
3.1	Calibrated and searched	144	–	–	–	–
3.1.1	**Archival extension**, reported	115	463	16	16	362
3.1.1.1	Class A, fine channels[a]	–	67	11	–	48
3.1.2	**Archival tail**, searched after the extension was frozen	8	32	5	5	23
3.1.2.1	Class A, fine channels[a]	–	14	5	–	10
3.1.3	ϵ Eri, both pointings[d]	8	32	–	–	–
3.1.4	No window reached the export	13	0	–	–	–
3.2	No calibration in the ALMA delivery	18	–	–	–	–
3.3	Calibration failed	11	–	–	–	–
3.4	Not attempted	3	–	–	–	–
3.5	Working disk insufficient	1	–	–	–	–
4	Beyond 40 pc: out-of-sample control	47	198	22	–	90
5	Rank-calibration sample[b]	326	1322	56	–	–

[a] The two classes partition the census windows, not its blocks: 255 blocks carry a Class A window and 392 a Class B one, and most carry both, so the hours of the two rows partition the windows and not the observing time.
[b] Not an out-of-sample set and not an addition to this ledger: every one of its blocks is counted in a row above, in the primary census or in the hold-out, and it is listed apart only to keep it from being counted twice.
[c] Panels 1 and 2 overlap rather than add: 479 of these 484 blocks are row 1.1, and the other 5 carry no member observing unit in the archive metadata, so the scope query does not expose them; 4 of the 5 are in the primary census.
[d] Every one is an ϵ Eri Band 6 block. The archival pointings put the star either at the centre of the field, at a response of 1.00, or well off it, where it is 0.22–0.27 over the 16 of this row's 32 windows that lie off axis, above the 0.22 retention floor of §2 in every case. All of them are searched, each from its own recorded phase centre, and their conventions are in Appendix A.
The same band's 12 windows that fall in the primary census are counted in row 2.1 and carry every census statistic; the on-axis pointings reach a response of 1.00.

tracted in apparent flux density from phase-rotated visibilities (Appendix A), and a noise rise toward the field edge would put the stellar rank *above* one half, the opposite of what is measured. The ensemble's own weak rise with radius is removed by standardising each probe within its radius bin (`documentation/appendix_B_spatial_normalisation.md` of the data archive, White & Dey 2026).

A calibration error that varied with frequency – a bandpass residual, or a coarse-mode system temperature interpolated near a telluric line (Hunter et al. 2018) – multiplies the continuum, so it would act at the star and not on empty sky, with the sign observed. *Its amplitude is too small.* A fractional error ϵ reaches the trigger only if $\epsilon = 5\sigma/S_{\rm c}$, with the continuum $S_{\rm c}$ measured for 1557 of the 1663 windows. No window requires as little as the 5–20 per cent a coarse system temperature leaves at an ozone line: the least anywhere is 21.4 per cent and the median 1073. The bandpass branch needs 21× the 1 per cent quoted for the worst bands (ALMA Science Portal 2026; Kameno 2014).

An additive visibility error (a correlator offset, crosstalk or coherent interference) images at the phase centre, where the star sits and the annulus does not. The excess is not confined there: the ratio is 1.12 (1.05–1.18) in the 332 windows with the star within one synthesised beam of the phase centre and 1.07 (0.95–1.18) in the 73 beyond it, with 20 and 6 unattributed crossings against 19.3 and 4.5 expected. The outer set (43 of its windows toward ALMA J153702653−33192492) is too small to exclude a contribution. The excess appears only in the 7 m-array data, 1.15 (1.09–1.22) against 1.04 (0.95–1.13) on the 12 m array, and its cause is not identified.

B.1. *The reserved hold-out, and what it calibrates*

The statistic and the mask are set on the searched sample and cannot calibrate themselves, so a further sample is held out by an identifier rule, $\mathrm{sha256}(\mathrm{UID})_{[:8]} \bmod 5 = 0$, with two guards using identifiers alone: a block on a named list stays in the survey, and no system loses every block. Its independence of the statistic and the mask is our assertion, not something a reader can check. 77 of 484 blocks are reserved, 315 windows toward 36 stars. Because the rule hashes the block and not the session, 64 per cent of the reserved set shares calibrators, weather and setup with a searched block – a session-level hash would have been stronger – so every quantity from it is also given over the 28 blocks that share no session.

B.2. *The recurrence statistic under the null*

The bound in the crossing ledger is one formula. Writing S and $\sigma_{\rm d}$ for the discovery amplitude and its error, and $\bar{a}$ and $\bar{\sigma} = (\sum_i \sigma_i^{-2})^{-1/2}$ for the inverse-variance combination of the repeat readings at the predicted channel,

$$\sigma_{\rm excl} = \frac{S - \bar{a}}{\sqrt{\sigma_{\rm d}^2 + \bar{\sigma}^2}} = \frac{T_{\rm pers} - T_{\rm rep}}{\sqrt{1 + (T_{\rm pers}/T_\star)^2}}, \qquad \text{(B1)}$$

with $T_{\rm pers} = S/\bar{\sigma}$ and $T_{\rm rep} = \bar{a}/\bar{\sigma}$, so every bound follows from Eq. B1 and three columns of that table. $T_{\rm rep}$ is one cell at the carried drift. The maximum over every drift

TABLE A2
The survey ledger, part two: events, dispositions and confirmation. Every headline count is read from here. Each sub-row is a subset of the row its number extends, and a row whose note names sub-rows is their sum.

Count	Value	What it counts
1 Threshold crossings at $T_\star \geq 5$	51	rows 1.1 and 1.2
1.1 … in the primary census	49	rows 1.1.1 and 1.1.2
1.1.1 Class A, fine channels	49	the carrier experiment
1.1.2 Class B, coarse channels	0	the released catalogue's 4, all in the one block that fails the quality criterion, fall below the trigger on recalibration
1.2 … in the archival tail	2	both unattributed, toward HD 14055, in windows no expectation covers (Table A1)
2 Quality-passing crossings of the census	49	the same rows as 1.1.1
2.1 … attributed to a transition	23	of 23 coincidences inside the mask in the star's own frame; rows 2.1.1 and 2.1.2
2.1.1 … with evidence beyond the coincidence	16	recurrence at the predicted cell, a second transition of the same species, or catalogued gas on the sightline
2.1.2 … on the frequency coincidence alone	7	η Crv, CD−57 1054, CP−72 2713, HD 161868, HIP 10679 and HR 1010: consistent with molecular emission, not identified as it
2.2 … unattributed	26	rows 2.2.1 and 2.2.2
2.2.1 … in the primary census	26	against 23.8 expected by chance at the 405 windows' own control positions – a count chance delivers with probability 0.36
2.2.2 … coincident with an atmospheric transition	0	within 62.5 MHz of one in the topocentric frame (Appendix E), and counted as atmospheric on both sides of that comparison
2.2.3 … recovered in a later block	0	no confirmed signal; 13 attributed ones do
3 Class A systems	61	of 83 surveyed; rows 3.1 and 3.2
3.1 … with same-frequency coverage in a separated session	**40**	**the only confirmation count quoted**: a later block in another session, among the census and the repeat blocks the recurrence test reads, covering a frequency the first covered; 13 of them over a year later. For an accelerating carrier, confirmation further requires the repeat to fall at a matching orbital phase
3.2 … with no separated second block at a covered frequency	21	confirmation impossible over the 142 blocks the recurrence test reads; 36 of all 83
4 Class A union bandwidth	47.7 GHz	26 disjoint islands
4.1 … removed by the line mask	6.4 %	3.05 GHz: the figure the text quotes, against 4.3 per cent of the 118.1 GHz archival union

TABLE A3
The 67 ALMA proposals the searched blocks belong to, as ADS/JAO.ALMA#*code*. Every census block belongs to one.

2012.1.00142.S, 2012.1.00198.S, 2012.1.00268.S, 2012.1.00385.S, 2012.1.00437.S, 2012.1.00993.S, 2013.1.00359.S, 2013.1.00588.S, 2013.1.00645.S, 2013.1.01147.S, 2013.1.01214.S, 2015.1.00307.S, 2015.1.00633.S, 2015.1.00783.S, 2015.1.01015.S, 2015.1.01260.S, 2016.1.00104.S, 2016.1.00358.S, 2016.1.00515.S, 2016.1.00637.S, 2016.1.00803.S, 2016.1.00817.S, 2016.1.00878.S, 2016.1.00880.S, 2016.2.00015.S, 2016.2.00200.S, 2016.A.00013.S, 2017.1.00167.S, 2017.1.00200.S, 2017.1.00215.S, 2017.1.00561.S, 2017.1.00698.S, 2017.1.00704.S, 2017.1.00786.S, 2017.1.00986.S, 2017.1.01583.S, 2017.1.01644.S, 2018.1.00045.S, 2018.1.00072.S, 2018.1.00126.S, 2018.1.00461.S, 2018.1.00470.S, 2018.1.00947.S, 2018.1.01149.S, 2018.1.01545.S, 2018.1.01833.S, 2019.1.00189.S, 2019.1.00696.S, 2019.1.01443.T, 2019.1.01517.S, 2019.1.01603.S, 2019.1.01681.S, 2019.2.00208.S, 2021.1.00485.S, 2021.1.01209.S, 2021.2.00097.S, 2022.1.00134.S, 2022.1.00338.L, 2022.1.00793.S, 2022.1.01163.S, 2023.1.00390.S, 2023.1.00487.S, 2024.1.00165.S, 2024.1.00595.S, 2024.1.00722.S, 2024.1.01095.S, 2024.A.00040.S

at that cell is tabulated beside it but never compared with $T_{\rm pers}$: it is positive on average whatever is there. As an amplitude selected from 10^2 to 10^7 cells at $T_\star \simeq 5$ is biased high, the limit is stated as $(T_{\rm rep}+5)/T_{\rm pers}$. The reading is centred on zero, scattering by 0.994 about −0.016 over 327 single-cell readings of every repeat block in the census and hold-out, less the 5 of the crossing that recurs, a molecular line. The bound's power is $\Phi(T_{\rm pers}-5)$ at the discovery amplitude and $\Phi(5T_{\rm pers}/T_\star-5)$ at the trigger amplitude $5S/T_\star$, the faintest that could have produced the crossing; the 5 where these are smallest are reported as untested. Each crossing's measured recovery chance accompanies this paper as data.

C. THE LINE MASK

A crossing that falls on a known molecular transition has an astrophysical explanation to hand, and the mask says which crossings those are.

The list, by one rule. The mask holds every transition of the 12 species these setups cover that are detected in discs or molecular clouds – the three CO isotopologues, HCN, HCO^+, CS, CN, SiO, H_2CO, SO, [C I] and hydrogen recombination – inside the 46 searched intervals below 150 K in upper-state energy, 111 transitions (Table C1), main hyperfine components only and no strength cut, from one Splatalogue query (Remijan et al. 2007) of CDMS and JPL (Pickett et al. 1998; Müller et al. 2005) dated 2026 October 6.

The windows were placed for the observers' own lines, and that is a limitation: 39 per cent of the 405 Class A windows are centred within $\pm 50\,{\rm km\,s^{-1}}$ of a listed transition, against 1–8 per cent for the same windows displaced, so *unattributed* means unattributed against a list these setups shaped. The list is not the tuning, though: in the 380 windows whose setup the archive records, the observers named the species in 270 of the 464 cases of a listed species inside a window, and in 0 of the 100 windows holding an SO transition. A wider list, every transition below 150 K of all 225 species with an interstellar or circumstellar detection, would cover 98 per cent of the Class A union

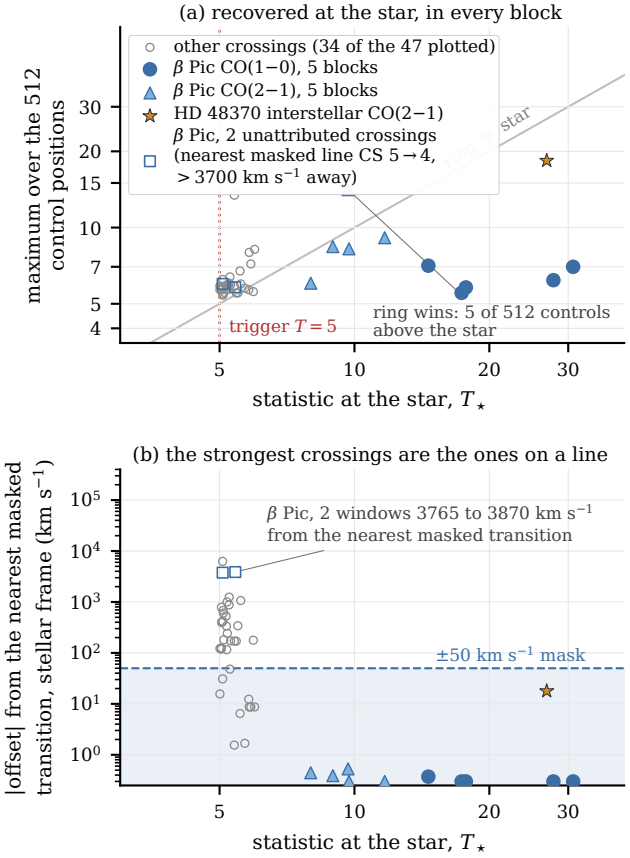


FIG. B1.— The β Pictoris positive control. *(a)* Stellar statistic against the maximum over the window's 512 control positions, filled for blocks recovering its CO. *(b)* Offset from the nearest masked transition in the star's own frame, $\pm 50\,\mathrm{km\,s^{-1}}$ marked: its 10 CO crossings lie 0.1–0.5 $\mathrm{km\,s^{-1}}$ from CO, its 2 unattributed ones 3765–3870 $\mathrm{km\,s^{-1}}$ from any; grey, the other quality-passing crossings.

and attribute every unattributed crossing and 100 per cent of the control exceedances alike, so at that density proximity is not an identification. The closest approach to one, the 2 unattributed crossings of β Pictoris 8.3 and 8.8 $\mathrm{km\,s^{-1}}$ from methanol $5_K \rightarrow 4_K$ components, fails three spectroscopic tests (`documentation/appendix_C_line_mask.md` of the data archive, White & Dey 2026).

Attribution is decided in the star's own rest frame. A crossing's frequency is carried from the topocentric value to the barycentre using its own block's pointing and mid-time, and then to the stellar rest frame using the star's systemic velocity; an offset $\Delta v_\star = c\,(\nu_\star - \nu_0)/\nu_0$ is positive for a crossing above the transition in frequency. *Every crossing is also read in the local standard of rest, where interstellar gas is at rest.* Toward ALMA J153702653−33192492, HD 139084 B, HD 285968 and HD 48370 the CO($1 \rightarrow 0$) survey of Dame & Thaddeus (2022); Dame et al. (2001) shows emission at the crossing's own LSR velocity (within 3 $\mathrm{km\,s^{-1}}$ of its peak, $\geq$0.23 K), so 6 of the 23 coincidences are interstellar; elsewhere the Planck CO map (Planck Collaboration et al. 2016) stays below 0.178 $\mathrm{K\,km\,s^{-1}}$. HD 48370's coincidence is $-17.8\,\mathrm{km\,s^{-1}}$ from the star's own CO($2 \rightarrow 1$) but at $+23.9\,\mathrm{km\,s^{-1}}$ in the LSR on a cloud line peaking at +21.8, so its extended emission is a cloud and not a disc.

TABLE C1
THE LINE MASK, BY SPECIES: THE NUMBER N OF TRANSITIONS, THE FREQUENCY RANGE THEY SPAN INSIDE THE SEARCHED INTERVALS AND THEIR UPPER-STATE ENERGIES.

Species	N	ν_0 (GHz)	E_u (K)
H	5	92.03 – 231.90	–
SO	15	100.03 – 682.68	16–143
H_2CO	13	101.33 – 491.97	10–106
$C^{18}O$	3	109.78 – 329.33	5–32
CN	50	113.12 – 680.30	5–114
CO	3	115.27 – 345.80	6–33
CS	3	195.95 – 342.88	24–66
SiO	2	217.10 – 347.33	31–75
^{13}CO	8	220.40 – 330.59	16–32
HCN	7	354.50 – 354.51	43–43
HCO^+	1	356.73 – 356.73	43
[C I]	1	492.16 – 492.16	24

Table C1 gives the mask species by species; the hydrogen recombination lines have no tabulated upper-state energy.

The coincidences. 23 of the 49 quality-passing census crossings fall within the mask in their own star's frame, and these are the 23 attributions. The three on sulphur monoxide (CD−57 1054, CP−72 2713 and HD 161868, at −1.7, −12.3 and $-48.0\,\mathrm{km\,s^{-1}}$) rest on the coincidence alone: in the LSR their offsets are −3.5, −12.8 and $-53.4\,\mathrm{km\,s^{-1}}$, none within the cloud test, and no CO($3 \rightarrow 2$) feature above the trigger accompanies any. The hold-out is treated identically, 1 attributed and 4 unattributed (§4.6.1).

The half-width, and what it costs. Keplerian motion where these molecules survive, $\lesssim$15 $\mathrm{km\,s^{-1}}$ beyond 10 au, sets it at $\pm 50\,\mathrm{km\,s^{-1}}$. *The cost is stated on the union of searched sky frequencies*: the frequencies within that half-width of a listed transition's rest frequency, merged where transitions overlap, are 3.05 GHz of the 47.7 GHz Class A union, 6.4 per cent, and 5.04 GHz of 118.1 GHz over both classes, 4.3 per cent; the wider list's share is measured the same way. A union over windows toward different stars has no single rest frame, so this describes the coverage and not any one star; a limit is formed instead over its own window's channels outside the mask in that star's own frame (§3.2).

Of the 6 labelled crossings not recovered (§4.6), one rests on a velocity instead, its sightline carrying interstellar emission 0.5 $\mathrm{km\,s^{-1}}$ from the crossing, with 192–491 of the 512 control positions above the star in every repeat block.

D. THE FALSE-ALARM EXPECTATION AND THE CONTROL ENSEMBLE

D.1. *The expectation, and what it is conditional on*

One attribution set serves both sides of the comparison: the mask of Appendix C, so a control maximum inside it is attributed exactly as a crossing there is. Each of the 405 Class A windows of the primary census contributes the fraction of its own 512 control positions that reaches the trigger – no cell count enters it and no form is assumed for the tail – and summed over the windows 36.5 are expected to trigger by chance. How many of those would be attributed is measured on the channels the control maxima land on, which exist only for the 8 control positions per window that retain a dynamic spectrum (§4.5): 3 192 positions in 399 windows, carrying 98.6 per cent of the expectation and indistinguishable

from their window's full ensemble in rank ($P = 0.88$) and in exceedance rate (+0.1 standard deviations). 92 of their 289 exceedances fall inside the mask, carried to the star's frame exactly as a stellar cell, so the attribution takes 32 per cent of the expectation where a uniform landing would take 23. The atmospheric screen (Appendix E) then takes 9 of 197, leaving 23.8 expected unattributed Class A crossings against 26 observed. Every control rate is measured on the data its window's statistic is read from, including the 28 Class A windows with recombined in-beam fields; on the first extraction the expectation would be 0.7 lower. The 14 Class A windows of the archival tail, with its two unattributed crossings toward HD 14055, and the 1 of HD 139084 B enter neither side.

A count is compared with the distribution of counts chance delivers. Over 200 000 resamplings – each window's rate and attributed share redrawn from its own probes, and the execution blocks resampled because windows share them – a count of 26 or more arises with probability 0.36, conditional on the control-position model and not on an independently established null. Added to each resampled rate, 12 further crossings still leave 26 credible at 0.058 and one more does not (0.042 against 5 per cent). Resampling blocks rather than windows widens the scatter by a factor 1.13, and a pseudo-star read at one control position in every window of a block gives a count no more dispersed than one read window by window (variance ratio 0.99) and reaches 26 with probability 0.35, so the Poisson step is not optimistic; these checks, the comparison by configuration and the phase-centre split are archived with the data (White & Dey 2026).

The one assumption the searched data cannot test is that a star behaves like a control position in its own field. The star carries 1.108 times a control position's cells in the band $4 \le T < 5$ (1.053–1.163 at 95 per cent, blocks resampled). Calibrating the stellar rate against that ratio instead, the sensitivity test of §4.3.1, lowers the bound to 9; the whole-beam conclusion is unchanged under it, and the control positions' own rate is adopted as the more demanding of the two for the count under test. The 26 would stand above chance at 5 per cent only if stars triggered at 0.76 times a control position's rate or less, and every ratio measured is an excess in the census. The reserved blocks trigger in 5 of 62 windows against 6.5 predicted, and the 28 sharing no session with a searched block hold one trigger event against 1.9 expected, so they neither confirm nor contradict the ratio (Appendix B.1).

Near a band edge the pooled exceedance rate is ×2.464 the interior rate, but the window toward HD 48370 contributes 59 per cent of every near-edge exceedance in the survey; without it the ratio is ×1.020, and on a held-out half ×1.181 ($p = 0.24$). No edge correction is therefore applied.

What the rank resolves. $1/(512 + 1) = 1.95 \times 10^{-3}$ is the control ensemble's *resolution*, not the probability that noise produced the window: a survey of 1663 windows would need a rank 65 times finer, and correlated controls inside the beam leave an annulus resolving no better than 0.023–0.032. No window can be significant on the rank alone. The widened recurrence window of §4.6 is counted in windows, a window covering several crossings being one measurement: the 194 pairings are 113 distinct windows, each one's chance formed from its independent channel–drift cells; with the rate taken from its own control positions instead the expectation is 10.0 ($P = 0.12$), and the larger P is quoted. The full treatment is in `documentation/appendix_D_false_alarm.md` of the data archive (White & Dey 2026).

D.2. *Execution-block quality*

The comparison assumes each block's control ensemble behaves like noise, and one block makes that false: all 512 of its probes exceed the trigger, the smallest at 6.6. Every block is therefore judged on the median of its windows' control ensembles divided by its class median, which also flags 5 of the 1663 windows individually at twice the survey median. *The criterion is post hoc*, written after the condition was seen. It is trustworthy because the failing block sits at 4.1 and the next at 1.12, a factor 3.7 apart, and all 41 thresholds between select the same block. 1 of 407 census blocks fails; its 4 coarse-channel windows hold no census crossing once ALMA's final flags are restored (§3).

D.3. *The registration of the predicted cell*

The recurrence test of §4.6 reads one channel at one drift, registered between epochs to 0.75 of a channel in every window, the frame correction moving it by up to 116 channels. Widening each reading to the worst cell within $0.45\,\mathrm{km\,s^{-1}}$, pinning it at every anchor along the discovery track, or raising the bar to 4.0 changes no disposition. Registration to a single channel is tested only on the 3 recovery readings of HD 285968, whose line is confined to one channel and registers to 0.06 of a channel; the other recoveries test the frame correction only to the width of a resolved line.

D.4. *From the widened window's exceedances to the pairs*

Two counts of §4.6 are counts of different objects. The 18 above-trigger readings are window readings, one per pairing of an unattributed crossing with a window covering its frequency in another block – 194 such pairings over the 28 crossings a later catalogued block covers – so a pair of unattributed crossings is read twice: 6 readings are second readings of 12 pairs, 2 outside the interval the pair test admits and 10 inside it, the set of Table 5, the other 1 of the 10 wanting a discovery spectrum or a fitted drift. The accounting for each pair is `candidates/drift_pair_accounting.csv` of the data archive. Of the 5 pairs whose epochs are close enough together for the predicted offset to be tighter than the searched window, every one fails a drift changing at a constant rate, by 3.6 times that resolution in the mildest case. The control row is a line that does not drift: the CO(2–1) line of HD 285968 is recovered at one stellar-frame frequency 5.0 days apart, within 0.6 of the comparison's resolution.

β Pictoris is the case the comparison cannot dismiss. Its two crossings, $106\,\mathrm{km\,s^{-1}}$ apart in the star's frame in blocks 24.4 h apart, miss a constant acceleration by 132 times the resolution, but the mean acceleration they need, $1.21\,\mathrm{m\,s^{-2}}$, is inside the $3.6\,\mathrm{m\,s^{-2}}$ ceiling and a circular orbit about the $1.75\,\mathrm{M_\odot}$ star (Crifo et al. 1997) reproduces all three numbers the pair fixes, so the pair is not separable (the orbit fit is in this appendix's full treatment in the data archive).

TABLE D1
The crossings whose fitted drift lies on an outermost trial of their window's own grid. Frequencies are in the star's rest frame, rates in the observatory frame.

Star	block	$\nu_\star$ (GHz)	$\dot\nu_{\rm obs}$ (Hz s^{-1})	trial	$\lvert a_{\rm los}\rvert$ (m s^{-2})
HD 161868	`X10ca8`	345.649	+4195	127/127	≥ 3.64
TRAPPIST−1	`X259b`	346.174	−4623	0/137	≥ 4.00

D.5. *The drift grid's edges*

The drift trials run from $-\dot\nu_{\rm max}$ in half-channel steps to the first trial at or beyond $+\dot\nu_{\rm max}$, so the searched interval is asymmetric by up to one step; 2 of the 52 crossings with a fitted drift, in the census, the archival tail and the hold-out, lie on an outermost trial (Table D1), toward HD 161868, TRAPPIST−1, and are reported as lower bounds, 2 of them carrying no amplitude bound, which is why the table reaches beyond §4.6's range.

E. RADIO-FREQUENCY INTERFERENCE SCREENING

Interference is the first alternative explanation for an unattributed crossing. An interferometer is a poor detector of it – a signal not in the far field at the phase centre is suppressed in the correlation by its rapidly varying geometric phase – though allocations exist across the searched bands (International Telecommunication Union 2024), and the documented hazard, spaceborne cloud radar at 94.0–94.1 GHz (Stephens et al. 2002; Wehr et al. 2023; Emerson 2004), overlaps 1 of the 1663 census windows, with 0 crossings and 0 rank-passing events. The population tests of Table E1 find none of the signatures interference leaves, but *no per-channel interference flag exists, and none can be recovered*, because the correlator flagging was not retained. The one concentration a reader will notice, 14 crossings between 344.2 and 346.4 GHz, is the carbon monoxide tuning: 8 of the 30 unattributed crossings the census and the reserved hold-out place in a searched window, against 7.0 ± 0.7 expected from where the windows were tuned. The image-sideband test is a bound, not a clearance, and 0 crossings lie where the receivers are double-sideband (`documentation/appendix_E_rfi.md` of the data archive, White & Dey 2026).

None of this substitutes for the ON/OFF cadence a dedicated search uses: these are statements about the population, not a clearance of any one crossing.

F. PER-SYSTEM SENSITIVITY

The limit Table F1 quotes for a system is that of its best window, a bound pair appearing under its primary's name. A window inside the attribution mask constrains nothing (§3.2) and cannot be a best window. Beside each limit are that window's centre frequency ν and channel width $\Delta\nu$, the system's merged coverage $\Delta\nu_{\rm cov}$, the percentage *dec.* by which atmospheric decorrelation leaves the limit optimistic, and whether blocks of different sessions cover a common frequency more than a day (d) or a year (yr) apart, over the census and the 142 blocks the recurrence test reads. A dagger marks the 6 of the 60 systems whose own best window was one of the 49 injected into; the other 54 limits are transferred from the class average and carry the dispersion observed between injected windows, ×0.80 to ×1.58, itself transferred for the 8 systems whose best window lies outside the injected range in band, channel width, array configuration or integration time. A section mark flags the 8 limits set in a block whose fields were recombined after the limits were computed, which are conservative (§4.1). Each system's limit at both ends of the calibration interval, with decorrelation and with the exact channel response divided out, and its probability of reaching 10^{15} W when its transferred windows draw their factors from the injected ones, are in `sensitivity/system_limits.csv` of the data archive; the full treatment is in `documentation/appendix_F_per_system.md` (White & Dey 2026).

TABLE E1
The interference tests, each of which could have failed. p is against the same peaks redrawn on their own channel grids.

Test	Population	Result	p
Clustering in sky frequency	53 catalogue crossings (narrow-field rows) with a released peak	15 within ± 50 km s^{-1} of a searched transition against 5.8 expected: they cluster on the survey's own lines	0.0002
Crowding on one another	28 unattributed, quality-passing, of the census and the archival tail	no crowding beyond chance over six separations from 1 to 300 MHz once the six are allowed for (the smallest single one $p = 0.23$), where the 22 attributed crossings crowd at every one	0.63
Atmospheric transitions	56 crossings of the census and the hold-out, against 570 telluric lines	none inside the 62.5 MHz screening radius; 0 inside ± 50 km s^{-1} against 1.0 expected	–
Fixed intermediate frequency	1453 census windows with a released peak frequency	peak positions within their own windows uniform, median 0.50	–
Cross-target occupancy	3027 extracted windows toward 133 stars, every block set of Table A1 together	569 cross-star coincidences within 5 MHz against 554 ± 13 expected; *below* the null over the 109 above-trigger windows, 4 against 5.2 ± 1.9	0.15
Image sideband	the 26 unattributed crossings lying in a catalogued block	17 have no catalogued transition in the image interval and 4 have one the same block observed with nothing bright enough to leak; for 5 the parent frequency was not observed	–

REFERENCES

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TABLE F1

Per-system sensitivity of the carrier search. For each of the 61 systems the fine-channel search covers, the $EIRP_{90}$ of its best window with its own primary-beam, annual-parallax and smearing retentions applied, ordered by the limit. Columns and daggers are defined in Appendix F; the notes below give the other marks.

System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.	System	$EIRP_{90}$ (10^{14} W)	ν (GHz)	$\Delta\nu$ (kHz)	$\Delta\nu_{cov}$ (GHz)	dec. (%)	conf.
LP 876-10	0.87	345.83	488	1.7	4*	d	HR 1010	19	230.53	977	2.8	1	d
τ Cet	1.2§	230.55	488	1.8	0.4*	d	LHS 1140	19	230.00	488	1.8	0.4*	d
AU Mic	1.2	230.56	488	3.5	3*	yr	HD 172555	21	340.94	488	5.3	200	–
HD 69830	1.2	230.06	488	3.8	1	d	L 836-122	22	346.00	488	1.7	1	–
61 Vir	1.3	345.79	488	1.8	5*	yr	HD 155555C	22†	346.01	488	1.7	43	–
ϵ Eri	1.4§	229.98	977	2.3	–	yr	κ CrB	22†	230.57	977	1.7	2*	–
TRAPPIST-1	1.6	346.58	488	1.7	5	d	CP−72 2713	24	345.12	488	5.8	64	d
β Pic	1.7‡	226.85	122	4.4	2*	yr	HD 14055	24	230.11	488	7.2	3	yr
AT Mic AB	1.7	346.04	488	1.7	3	–	TWA 7	24	345.99	244	0.87	26*	d
GJ 3379	2.2	229.96	488	1.9	0.4*	d	GJ 2006A	24§	346.02	488	1.7	1	–
GJ 14	3.1	227.22	488	3.5	19	d	CD−57 1054	24	345.99	488	1.7	20	d
HN Lib	3.9†	229.99	488	1.8	0.4*	d	Wolf 219	26§	230.13	488	1.7	2*	d
HD 10647	4.4§	345.79	488	6.1	0.8	yr	LP 736-15	28	345.82	977	1.7	4*	–
η Crv	4.5	343.35	488	7.2	27	yr	TYC 2211-1309-1	29	346.00	488	1.7	5	–
HD 202628	5.4	230.51	488	1.8	4*	yr	UCAC3 176-23654	29	346.02	488	1.7	4	–
HD 53143	5.4	230.53	977	1.8	2*	yr	PM J03433+1958	31	230.13	488	1.7	2*	–
HD 207129	6.4§	230.57	488	1.8	3*	yr	UCAC2 20312880	33	345.96	488	1.7	11	–
GJ 849	6.8	230.02	488	1.9	0.4*	d	LP 353-51	33	346.02	488	1.7	26	–
TWA 3A	7.6†	230.54	122	0.44	35*	d	Barta 161 12	34	346.02	488	1.7	4*	–
HD 139664	7.8§	230.55	488	1.7	3*	–	HD 23484	38	230.53	977	1.8	0.2	d
HD 92945	8.3	355.11	488	3.7	6*	d	51 Eri	42	230.53	1953	1.7	11*	–
TX PsA	8.5	346.03	488	1.7	7	–	HD 139084	54	346.01	488	1.7	3	–
HD 285968	9.1	229.97	488	1.8	0.4*	d	WD 0407+179	84	230.13	488	1.7	2*	–
HD 161868	9.9	227.22	488	7.4	5	d	HD 48370	88†	230.73	122	3.3	0.4*	yr
HD 110411	13	219.58	244	4.7	2*	yr	HD 45184	93	230.53	977	1.7	2	d
HD 38858	13§	230.50	488	1.8	3*	yr	ALMA J153702653-33192492	99	345.98	488	14	5*	d
HIP 490	13†	230.53	244	0.86	3*	–	HD 31392	130	230.52	977	1.7	0.9	d
HD 107146	14	355.14	488	3.7	6*	d	HD 89452	260	230.53	977	1.7	0.7	d
LP 476-207	15	346.02	488	1.7	4	–	HD 377	400	230.54	977	1.7	68	d
HD 170773	17	230.13	488	7.4	2*	d	HIP 10679	–	–	–	0.11	–	–
HD 61005	19	230.52	244	19	3*	d							

Notes. A double dagger marks a row keeping less than half its channels outside the mask; HIP 10679, whose 2 windows are all inside it, is tabulated with dashes. *dec.* is measured where the delivery records a phase rms and imputed where it does not: an asterisk marks the 32 systems whose best window lies in a block with no phase record of its own, imputed from blocks matched to it within a factor of two in baseline length, against 27 measured, and a dash the one system whose blocks offer neither. The 3 per cent of §4.1 is therefore a mixture, 4 per cent over the measured systems and 3 over the imputed. The imputed blocks are not a random subset: 54 per cent are 12 m against 26 per cent of those with a record, that array's long baselines being what loses coherence. Their windows lose 1.9 per cent in the median, and 21 per cent had each performed like the ninetieth-percentile block matched to it, a condition on a calibration performance the archive does not report. Every value in the column is a floor: the phase rms is recorded on the calibrator, not on the target.

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